

Value Premium and Equity Term Structures of Value and Growth Firms ^{*}

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Abstract

This paper studies the impact of information processing and learning about expected future cashflows on the level and timing of risk premiums in the cross-section of firms. Learning with information sources of different qualities endogenously generates value firms, growth firms, and a value premium in equilibrium. Furthermore, the learning model predicts an upward-sloping equity term structure for value firms, and a flat equity term structure for growth firms. Using earnings, return, and news data on value and growth firms, we show that the predictions of the learning model are consistent with the data.

Keywords: Asset Pricing, Information Acquisition, Learning, Value Premium, Equity Term Structures of Value and Growth Firms

JEL Classification. D51, D53, D83, G12

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1 Introduction

The value premium, defined as the spread between the average returns on high book-to-market (value) and low book-to-market (growth) stocks, has arguably been one of the most studied cross-sectional phenomenon in finance (Fama and French, 1992, 1993). Recent research further explores the timing aspect of the value premium, studying the slope of the equity term structure in the cross-section of stock returns (Miller, 2020; Baele, Driessen, and Jankauskas, 2021; Gormsen and Lazarus, 2023; Giglio, Kelly, and Kozak, 2024). This evolving understanding of equity term structures, going beyond the focus on the aggregate market index, emphasizes the importance for asset pricing models to address both the average risk premium of value versus growth portfolios and the term structure of their discount rates.

In this paper, we study the role of investors’ information processing and learning in explaining the cross-section of expected returns and equity term structures. Unable to observe the driving components of output growth and firms’ cashflow growth, investors process the observable data to form and update their beliefs about future growth prospects. We emphasize how heterogeneous levels of news signal precision across firms can drive a wedge between their valuations and risk premiums. High signal precision enables investors to differentiate between priced and unpriced shocks, while low precision prompts reliance on aggregate information for updating beliefs about future firm performance. In equilibrium, learning with information sources of different qualities endogenously generates value firms, growth firms, a value premium, and equity term structures for value and growth firms having different shapes.

The learning model yields a sizable value premium of about 4%, a markedly upward-sloping equity term structure for value firms, and a flat equity term structure for growth firms. In addition, the model generates a realistic market risk premium of about 7%, and a moderately upward-sloping term structure of market risk premium.¹

¹Whether the equity term structure of the market is upward-sloping or downward-sloping in the data is the subject of an active debate. See, for instance, van Binsbergen, Brandt, and Koijen (2012), van Binsbergen, Hueskes, Koijen, and Vrugt (2013), van Binsbergen and Koijen (2017), Weber (2018), Gonçalves (2021a), Gonçalves (2021b), Gormsen (2021), and Bansal, Miller, Song, and Yaron (2021), Gormsen and Lazarus (2023), and Boguth, Carlson, Fisher, and Simutin (2023). Similarly, the literature offers little consensus on the empirical slope of the equity term structure in the cross-section of stock returns. Gormsen and Lazarus (2023) analyze risk-adjusted returns from single-stock dividend futures and find that these returns decline with maturity but do not systematically differ across firms. In contrast, Giglio et al. (2024) estimate an affine pricing model using a broad cross-section of equity returns, finding that the term structure of the value premium is upward-sloping, though the economic magnitude is small and statistically insignificant. Miller (2020), using no-arbitrage prices of continuation value claims, reports that value firms have higher and more steeply upward-sloping term structures. Meanwhile, Baele et al. (2021), using the implied cost of capital approach, document an upward-sloping term structure for growth firms but a downward-sloping one for value firms.

In the model, the output of the economy is driven by a time-varying expected growth rate. This expected growth rate is unobservable, and therefore needs to be estimated by the representative agent using observed changes in output.² Because the expected growth rate is mean-reverting and persistent, output growth risk increases with the time horizon in the model, as in the data. There are also two firms, V and G , in the economy. The firms' expected cashflow growth rates depend on the expected output growth rate as well as on unobservable firm-specific expected growth components. The agent needs to estimate the firm-specific components by observing the firms' realized cashflows as well as two information signals, one for firm V and one for firm G .

The parameters of the model are estimated by maximum likelihood using data on U.S. GDP and GDP forecasts, earnings and earnings forecasts on a portfolio of value firms, and earnings and earnings forecast on a portfolio of growth firms. The estimated signal precision of firm V is relatively low, whereas that of firm G is high. In addition, the correlation between the firm-specific expected growth component and the realized growth rate is slightly negative for firm V , and substantially negative for firm G . In the model, these features imply that firm V has lower valuations than firm G . That is, firms V and G endogenously emerge as value and growth firms, respectively. The model-implied risk premium of firm V is about 4% larger than that of firm G . That is, the model-implied value premium is about 4%, in line with the data. In addition, the model-implied term structure of risk premium is markedly upward-sloping for the value firm V , and flat for the growth firm G .

The economic mechanism underlying the model's results on risk premiums and valuations is as follows: The precision of the information signal affects how the agent discerns priced versus unpriced shocks. Indeed, a precise information signal on the firm-specific expected growth component implies that the agent is close to observing it. Consequently, the agent does not require a large risk premium to hold the firm's stock, and the firm's valuations are high. Conversely, if the information signal is imprecise, the agent is not able to clearly discern the value of the firm-specific expected growth component and turns to aggregate information, specifically priced shocks in output and cashflow growth, to update their expectations. Therefore, the agent requires a large risk premium and low valuations to hold the firm's stock. This explains why firm V , the firms with a low signal precision, has lower valuations and a higher risk premium than firm G , the firm with a high signal precision.

Each economic fundamental has a distinct impact on the firms' term structure of risk premium. The expected output growth component has a strong upward-sloping effect because it is persistent (Bansal and Yaron, 2004). The effect of the firm-specific expected growth

²See Detemple (1986, 1991), Veronesi (1999, 2000), Brennan and Xia (2001), Bansal and Yaron (2004), and Berrada (2006, 2009) among others.

component depends on the quality of information the agent receives about the firm. When the information signal is imprecise, the agent updates her beliefs about the firm-specific expected growth component using almost exclusively information provided by realized cashflows. Consequently, the agent's estimate is highly correlated to realized cashflow shocks. These shocks represent a priced source of risk that contribute to the firm's risk premium. As a negative shock hits the firm's realized cashflow, the agent updates her beliefs about the firm-specific expected future cashflow component downward. That is, the agent perceives long-run cashflow risks as being larger than short-term cashflow risks. Because these risks are priced, they imply an upward-sloping effect on the firm's term structure of risk premium. This explains why the value firm V has a markedly upward-sloping term structure of risk premium, implied by the upward-sloping effects of both the expected output growth component and the firm-specific expected growth component. On the other hand, when the information signal is precise, the agent updates her beliefs about the firm-specific expected growth component using mainly information provided by the signal. Since the signal is uncorrelated to the output, signal shocks are not priced, and hence the effect of the firm-specific expected growth component on the term structure of risk premium is weak. As a result, the term structure of risk premium is shaped principally by the effect of the expected output growth component. This explains why the growth firm G has a flatter term structure of risk premium than the value firm V . Adding the downward-sloping effect implied by the negative correlation between the firm-specific expected growth component and the realized growth rate yields a largely flat equity term structure for the growth firm G .

In summary, partial information and learning about firms' expected cashflows using information signals of different qualities endogenously generates value and growth firms, a value premium (Fama and French, 1993), a flat equity term structure for growth firms, and a markedly upward-sloping equity term structure for value firms.

To further validate the learning mechanism, we present consistent empirical evidence of cross-sectional differences in information precision. Specifically, we find that the response of earnings forecasts to output shocks is stronger for value firms than for growth firms, indicating a greater reliance on aggregate information when updating beliefs about value firms. Moreover, earnings forecasts of growth firms are subject to more volatile shocks, suggesting beliefs about growth firms are updated more aggressively than about value firms, consistent with the processing of more precise signals. We also observe that value firms typically receive less analyst and news coverage than growth firms, there is more analyst disagreement about value firms, and the errors in forecasting their earnings are larger. This suggests cross-sectional heterogeneity in precision of information, aligning with our model's mechanism.

Additionally, we contrast our findings with those of an equivalent full-information economy to show that our results are a direct consequence of the learning mechanism. In the full-information economy, the agent observes all economic variables and therefore does not need to learn. The results obtained in this full-information economy are strikingly different from those obtained in the partial-information economy with learning. Indeed, firm G has lower valuations and a lower risk premium than firm V in the full-information economy. That is, firm G endogenously emerges as the *value firm*, whereas firm V emerges as the *growth firm*. The value premium is consequently negative, equal to about -2% , in the full-information economy. Furthermore, the equity term structure of the value firm G is markedly downward-sloping, whereas that of the growth firm V is moderately downward-sloping. These results are at odds with key empirical patterns. This shows that partial information and learning are crucial, required features of the model.

This paper is related to the literature on the value premium. In [Zhang \(2005\)](#), costly investment reversibility represents a form of operating inflexibility that rationalizes the value premium. [Yogo \(2006\)](#) shows that utility derived from both nondurable and durable consumption goods can explain the value premium. In [Bansal, Dittmar, and Lundblad \(2005\)](#) and [Hansen, Heaton, and Li \(2008\)](#), the value premium results from the value firms' higher dividend exposure to aggregate expected growth relative to growth firms. [Santos and Veronesi \(2006\)](#) rationalize the value premium via the existence of fluctuations in both wages and dividends. [Lettau and Wachter \(2007, 2011\)](#) show that the value premium can be explained by the duration of assets. In [Kogan and Papanicolaou \(2014\)](#), the value premium is explained by the firms' different sensitivities to investment-specific technology shocks. [Tsai and Wachter \(2015\)](#) provide an explanation for the value premium based on rare events, highlighting the asymmetrical pricing of boom versus disaster risk. In their model, cashflows of growth stocks face greater exposure to boom risk compared to value stocks. However, the recent empirical findings of [Golubov and Konstantinidi \(2019\)](#) empirically question whether the value premium can be explained by operating inflexibility, cashflow risk, asset duration (see also [Chen, 2017](#)), and exposure to investment-specific technology shocks. They conclude that the value premium remains a puzzle.

We contribute to this literature by providing an alternative economic channel for the value premium, namely information processing and learning. In our model, more accurate information on a firm's expected cashflow yields lower risk, and therefore higher valuations and a lower risk premium. That is, the learning channel implies that firms endogenously emerging as growth firms in equilibrium have a lower risk premium than those endogenously emerging as value firms. Our mechanism relates to [Hansen et al. \(2008\)](#) in that perceived risk differences between value and growth firms stem from their exposure to aggregate growth,

but in our model, these differences arise endogenously from investor learning rather than from firms’ intrinsic cash flow dynamics. Similarly, our model generates an effect resembling the duration-based explanation of [Lettau and Wachter \(2007\)](#), but through the lens of investor perceptions: learning causes value firms’ perceived (posterior expected) dividend growth to be more correlated with long-term aggregate growth than that of growth firms. This effect emerges only when investors must infer firm fundamentals over time and does not appear in a full-information setting. By leveraging analyst expectation data, we provide novel insights into how information frictions shape the risk premiums of value and growth stocks.

Our paper is also related to equity term structure literature ([van Binsbergen et al., 2012, 2013](#); [van Binsbergen and Koijen, 2017](#); [Weber, 2018](#); [Gonçalves, 2021a,b](#); [Gormsen, 2021](#); [Bansal et al., 2021](#); [Gormsen and Lazarus, 2023](#); [Boguth et al., 2023](#)). [Belo, Collin-Dufresne, and Goldstein \(2015\)](#) shows that, when dividend dynamics are such that leverage ratios are stationary, the term structure of market risk premium is downward sloping. In [Croce, Lettau, and Ludvigson \(2015\)](#), investors’ limited information and bounded rationality generate a high market risk premium together with a decreasing term structure of market risk premium. [Hasler and Marfè \(2016\)](#) show that recoveries following rare economic disasters yield a downward-sloping term structure of market risk premium. [Marfè \(2017\)](#) obtains a decreasing term structure of market risk premium by investigating the impact of labor on dividend payouts. [Ai, Croce, Diercks, and Li \(2018\)](#) show that, when investment responds positively to contemporaneous productivity shocks and negatively to news about long-term productivity shocks, the term structure of market risk premium is downward sloping. [Hasler and Khapko \(2023\)](#) show that time-varying correlation between expected and realized cash-flow shocks produces a term structure of market risk premium that is moderately downward sloping unconditionally, markedly downward sloping in good times, and markedly upward sloping in bad times.

Our results contribute to this literature by showing that information processing and learning yield a sizable value premium, increasing equity term structures for value firms, and flat equity term structures for growth firms. Investor learning amplifies the perceived exposure of value firms to long-run cash flow risks, creating an upward-sloping term structure effect similar to long-duration-based mechanisms ([Lettau and Wachter, 2007](#)). At the same time, the persistence of expected output growth, as in long-run risk models ([Bansal and Yaron, 2004](#); [Bansal et al., 2021](#)), strengthens this effect. Unlike these frameworks, our mechanism comes from belief updating, showing how information frictions shape risk premiums across maturities.

The remainder of the paper is organized as follows. [Section 2](#) describes economic fundamentals and the learning problem, and solves for equilibrium asset prices; [Section 3](#) describes

the data used to estimate the model; Section 4 estimates the parameters of the model and presents the results; Section 5 concludes. Derivations and supplementary material are provided in Appendix.

2 The Model

This section describes the economy and solves for equilibrium asset prices.

2.1 Preferences and Cashflows

We consider a pure-exchange economy (Lucas, 1978) with a representative agent having recursive preferences (Kreps and Porteus, 1979; Epstein and Zin, 1989; Weil, 1989; Duffie and Epstein, 1992). These preferences allow for the separation between the elasticity of intertemporal substitution and the coefficient of relative risk aversion. Given a consumption process C , the utility at time t is defined as

$$U_t \equiv \left[(1 - \delta^\Delta) C_t^{\frac{1-\gamma}{\theta}} + \delta^\Delta \mathbb{E}_t [U_{t+\Delta}^{1-\gamma}]^{\frac{1}{\theta}} \right]^{\frac{\theta}{1-\gamma}},$$

where δ is the time discount factor, γ is the coefficient of risk aversion, ψ is the elasticity of intertemporal substitution, $\theta = \frac{1-\gamma}{1-\frac{1}{\psi}}$, and Δ is a time interval.

In equilibrium, the representative agent's consumption is equal to the output of the economy. The output C follows the standard dynamics considered in the long-run risk literature (Bansal and Yaron, 2004):

$$d \log C_t \equiv dy_t = (\mu + x_t)dt + \sigma_C dB_{y,t}, \quad (1)$$

where μ is the mean output growth, and σ_C is the conditional volatility. The expected output growth component x reverts towards a mean of zero with the following dynamics

$$dx_t = -\lambda_x x_t dt + \sigma_x dB_{x,t}, \quad (2)$$

where λ_x is the mean-reversion speed, and σ_x is the conditional volatility. The Brownian motions B_y , and B_x are assumed to be independent. Solving the stochastic differential equation (SDE) (1) yields

$$\log \left(\frac{C_T}{C_t} \right) = y_T - y_t = \mu(T - t) + \int_t^T x_u du + \sigma_C (B_{y,T} - B_{y,t}). \quad (3)$$

Equation (3) shows that the log output growth rate depends on the time integral of the process x . Therefore, shocks in x accumulate over time and permanently affect the future levels of output. In this case, the term structure of output growth risk is upward sloping—risk is concentrated in the long run—because of the accumulation of x shocks in y .

There are two firms in the economy: V and G .³ As in [Abel \(1999\)](#), cashflows paid by firm $i \in \{V, G\}$ are modeled as levered consumption

$$d \log D_{i,t} \equiv dy_{i,t} = (\mu + x_t + x_{i,t}) dt + \sigma_D dB_{y,t}, \quad i \in \{V, G\} \quad (4)$$

where σ_D is the conditional volatility of cashflows. The firm-specific expected growth component x_i reverts towards a mean of zero according to

$$dx_{i,t} = -\bar{\lambda}x_{i,t}dt + \bar{\sigma} \left(\rho_i dB_{y,t} + \sqrt{1 - \rho_i^2} dB_{x_i,t} \right), \quad i \in \{V, G\} \quad (5)$$

where $\bar{\lambda}$ is the mean-reversion speed, $\bar{\sigma}$ is the conditional volatility, and ρ_i is the conditional correlation between shocks in x_i and shocks in D_i . The Brownian motion B_{x_i} is assumed to be independent of both B_y and B_x .

It is worth mentioning that the output and cashflow dynamics provided in Equations (1) and (4) imply that firms V and G cashflow-to-output ratios are not mean-reverting processes. Yet, these dynamics together with the parameter values provided in [Section 4.1](#) imply that most of the simulated paths of the cashflow-to-output ratios feature a mean-reverting behavior. See [Appendix D](#) for further details.

The agent does not observe the expected growth components x and x_i . Therefore, she needs to filter them out by observing cashflows C and D_i , as well as a news signal on B_{x_i} ([Dumas, Kurshev, and Uppal, 2009](#); [Xiong and Yan, 2010](#)) satisfying

$$ds_{i,t} = p_i dB_{x_i,t} + \sqrt{1 - p_i^2} dB_{s_i,t}, \quad i \in \{V, G\}.$$

The parameter p_i denotes the precision of the signal, and B_{s_i} is an independent Brownian motion. The news signal provides information on shocks driving the firm-specific expected growth component x_i , and therefore on the firm's future cashflows. As detailed in [Section 4.2](#), information processing and learning endogenously produce, as an equilibrium outcome, firms that can be identified as either growth (high valuations) or value (low valuations).

It's important to note that, to isolate the impact of learning, we assume that all parame-

³We choose to designate the firms as V and G to align with our later estimation of the model parameters using value and growth portfolios. At this stage, however, we make no assumptions about the specific parameters governing the cashflow processes for these two firms.

ters affecting the two firms in our economy are identical, with two crucial exceptions. First, the two firms are differentiated mainly by how informative the news signals are about firm-specific shocks affecting expected growth. The informativeness of these signals is measured by their precision p_i . Specifically, when $p_i = 1$, the signal precisely reveals the firm-specific shocks, as $ds_{i,t} = dB_{x_i,t}$. In contrast, a precision of $p_i = 0$ renders the signal about the firm completely uninformative, $ds_{i,t} = dB_{s_i,t}$.

The second element of cross-sectional heterogeneity is the parameter ρ_i , which governs the correlation between shocks to a firm's expected growth and the realized cashflow shocks affecting both firms. Our initial analysis in Section 4.2.1 sets $\rho_i = 0$ for both firms, effectively isolating the effect of heterogeneity in signal precision only. When $\rho_i \neq 0$ in Section 4.2.2, the model's learning mechanism is enhanced. In this case, investors' ability to accurately learn about growth prospects of firm i becomes increasingly important due to the more intricate nature of the firm-specific growth component x_i . The presence of a correlation between shocks to x_i and realized cashflow shocks introduces a richer layer of information, amplifying the importance of precise news signals.

2.2 Learning

The expected growth rates of output and firm i cashflows vary over time due to shocks that come from two sources: the output growth component x defined in (2) and the firm-specific component x_i defined in (5). The agent observes the realized paths of output, firm i cashflows, and the news signal but does not observe the expected growth components x and x_i . Therefore, the agent needs to filter out, using her observation filtration $\mathbf{F}^o = \{\mathcal{F}_t^o\}_{t \geq 0}$ and Bayesian updating rules, the unobservable components x and x_i . The sigma-algebra $\mathcal{F}_t^o = \sigma(C_u, D_{V,u}, D_{G,u}, s_{V,u}, s_{G,u} : 0 \leq u \leq t)$ defining the agent's observation filtration is generated by the output C , firm V cashflow D_V , firm G cashflow D_G , firm V news signal s_V , and firm G news signal s_G .

Proposition 1 provides the dynamics of output C and the filter $\hat{x}$ —the agent's estimate of x —under the agent's observation filtration.

Proposition 1. *Under the observation filtration, the dynamics of output C_t and the filter $\hat{x}_t \equiv \mathbb{E}[x_t | \mathcal{F}_t^o]$ satisfy*

$$d \log C_t = (\mu + \hat{x}_t)dt + \sigma_C d\hat{B}_t, \quad (6)$$

$$d\hat{x}_t = -\lambda_x \hat{x}_t dt + \hat{\sigma}_x d\hat{B}_t, \quad (7)$$

where $\hat{B}_t$ is a Brownian motion under the agent's observation filtration, and $\hat{\sigma}_x \equiv \frac{\gamma_x}{\sigma_C}$. γ_x is

the steady-state posterior variance of x provided in Appendix A.1.

Proof. See Appendix A.1.

Observing the process of output (1) allows the representative agent to learn about its unobservable driving force, x_t . Equation (7) describes the updating rule for the agent's *expectations* of the unobservable expected growth component x , which is called the filter $\hat{x}_t$. The learning exercise implies that the filter $\hat{x}_t$ is perfectly correlated with the output. This occurs because the only source of information the agent has to infer the unobservable expected growth component is the observation of the output. Indeed, because firm cashflow shocks coincide with output shocks, cashflows do not provide information on x that is not already provided by the output. In other words, the observation of cashflows is redundant when learning about x .

Because the agent does not observe the firm-specific expected growth component x_i , she needs to filter it out by observing the output, cashflows, and the news signal. Proposition 2 characterizes the dynamics of firm i cashflows D_i and the filter $\hat{x}_i$ —the agent's estimate of x_i —under the agent's observation filtration.

Proposition 2. *Under the observation filtration, the dynamics of firm i cashflows D_i and the filter $\hat{x}_{i,t} \equiv \mathbb{E}[x_{i,t} | \mathcal{F}_t^o]$ satisfy*

$$\begin{aligned} d \log D_{i,t} &= (\mu + \hat{x}_t + \hat{x}_{i,t}) dt + \sigma_D d\hat{B}_t, \\ d\hat{x}_{i,t} &= -\bar{\lambda} \hat{x}_{i,t} dt + \hat{\sigma}_{x_i} d\hat{B}_t + p_i \bar{\sigma} \sqrt{1 - \rho_i^2} ds_{i,t}, \quad i \in \{V, G\} \end{aligned} \quad (8)$$

where $\hat{x}$ is defined in Proposition 1, $\hat{\sigma}_{x_i} \equiv \frac{\rho_i \bar{\sigma} (\sigma_D - \sigma_C) + \gamma_{x_i}}{\sigma_D - \sigma_C}$, and γ_{x_i} is the steady-state posterior variance of x_i provided in Appendix A.2.

Proof. See Appendix A.2.

The dynamics in (8) show that the agent updates the filter $\hat{x}_i$ using two pieces of information. The first is the cashflow growth surprise $d\hat{B}$, and the second is the news signal shock ds_i . Consequently, the vector $[\hat{\sigma}_{x_i} \ p_i \bar{\sigma} \sqrt{1 - \rho_i^2}]$ represents the weights assigned by the investor to each source of information. These weights vary depending on p_i . As the precision of the news signal changes, the weights adjust in opposite directions. When the precision of the signal is equal to zero, the loading on the signal is unsurprisingly equal to zero too. Therefore, the filter is solely driven by cashflow surprises when $p_i = 0$; the filter and cashflows become perfectly correlated. As the precision of the signal increases, the loading on the news signal rises, resulting in a reduced weight of cashflow surprises in the filter update.

Corollary 1. *As the news signal precision p_i increases, $\hat{\sigma}_{x_i}$ in Equation (8) decreases, indicating a reduced weight the agent assigns to priced shocks when updating the filter $\hat{x}_i$.*

$\hat{\sigma}_{x_i}$ captures the conditional covariance between updates in expectations about future firm i cashflows, $d\hat{x}_{i,t}$, and realized cashflow growth surprises, $d\hat{B}$, which are priced shocks in equilibrium:

$$\frac{1}{dt} \text{Cov}_t(d\hat{x}_{i,t}, d\hat{B}_t) = \hat{\sigma}_{x_i}.$$

As Corollary 1 establishes, this covariance is inversely related to the precision of the news signal. When the news signal is imprecise, investors view shocks to firm-specific expected growth component as closely linked to priced shocks. However, as the signal precision improves, this perceived link weakens. With highly accurate signals, investors are better equipped to distinguish priced and unpriced shocks. When the signal is perfectly precise ($p_i = 1$) and there is no correlation in cashflow shocks ($\rho_i = 0$), $\hat{\sigma}_{x_i}$ decreases to zero, indicating a complete separation of firm-specific information from market-wide cashflow growth surprises.

In addition to influencing the weight investor assigns to priced shocks, signal precision also affects the volatility of their beliefs.

Corollary 2. *As the news signal precision p_i increases, beliefs about firm-specific expected growth become more volatile, provided that cashflows vary more than output ($\sigma_D > \sigma_C$).*

This corollary follows directly from the conditional variance of the filter in Equation (8),

$$\frac{1}{dt} \text{Var}(d\hat{x}_{i,t}) = \hat{\sigma}_{x_i}^2 + p_i^2 \bar{\sigma}^2 (1 - \rho_i^2),$$

which increases with signal precision if $\sigma_D > \sigma_C$. Intuitively, when a firm's signal becomes more precise, it injects additional information into the updating of $\hat{x}_{i,t}$. Consequently, the agent adjusts her belief regarding the firm's expected growth more aggressively, resulting in higher instantaneous volatility.

Corollaries 1 and 2 suggest the following insights: for a firm with higher signal precision, the agent perceives its expected growth as more volatile. At the same time, the agent can discern that most of this volatility is idiosyncratic and not priced.

2.3 Asset Pricing

Recursive preferences lead to a non-affine state-price density. Therefore, to solve for asset prices and preserve analytic tractability, we follow the methodology presented by [Eraker and](#)

Shaliastovich (2008), which is based on Campbell and Shiller (1988) log-linearization.⁴

Proposition 3 characterizes the state-price density, the risk-free rate, and the price of risk in the economy.

Proposition 3. *The equilibrium state-price density has dynamics given by*

$$\frac{dM_t}{M_t} = -r_t dt - \Lambda d\hat{B}_t,$$

where the risk-free rate satisfies

$$r_t = r_0 + \frac{1}{\psi} \hat{x}_t,$$

with

$$r_0 = -\frac{1-\gamma}{1-1/\psi} \log \delta + \frac{1/\psi - \gamma}{1-1/\psi} \log k_1 + \gamma\mu - \frac{1}{2} \Theta(\hat{\sigma}_x)$$

and $\Theta(\hat{\sigma}_x)$ defined in Appendix A.5.

The market price of risk satisfies

$$\Lambda = \gamma\sigma_C + \frac{\gamma - 1/\psi}{1/k_1 - (1 - \lambda_x)} \hat{\sigma}_x,$$

where $\hat{\sigma}_x \equiv \gamma_x / \sigma_C$.

Proof. See Appendix A.5.

The risk-free rate increases with $\hat{x}$, irrespective of the value of the parameters. The reason is that an increase in $\hat{x}$ implies an increase in expected consumption growth, as can be seen in equation (6). Because the investor smoothes consumption over time, an increase in expected future consumption yields an increase in current consumption and therefore a decrease in risk-free investments. As a result, the risk-free rate increases when $\hat{x}$ increases.

The market price of risk is the sum of two components. If the investor has CRRA preferences, $\gamma = 1/\psi$ and only the first component of the market price of risk remains. This component corresponds to the reward for bearing consumption growth risk (Lucas, 1978). The second component rewards for bearing expected consumption growth risk associated to variations in $\hat{x}$. When the agent has a preference for early resolution of uncertainty ($\gamma > 1/\psi$), the second component is positive (Bansal and Yaron, 2004). Conversely, a

⁴Campbell, Lo, and MacKinlay (1997) and Bansal, Kiku, and Yaron (2012) show that the log-linearization yields low approximation errors in a Gaussian setting. Hasler and Marfè (2016) document that it is also accurate when state variables follow jump-diffusions. For notational ease, we will treat the approximated results as exact in what follows.

preference for late resolution of uncertainty ($\gamma < 1/\psi$) results in a negative value for this component.

Proposition 4 characterizes dividend strip prices and risk premiums for the two firms in our model. In our setting, the market pays the sum of the cashflows from both firms, $D_{m,t} = D_{V,t} + D_{G,t}$. Consequently, the time- t price of a market dividend strip with maturity T , or equivalently time to maturity $\tau \equiv T - t$, is the sum of the two firms' dividend strips: $S_{m,t}(\tau) = S_{V,t}(\tau) + S_{G,t}(\tau)$. Similarly, the stock market price is the sum of the two firms' stock prices: $P_{m,t} = P_{V,t} + P_{G,t}$.

Proposition 4. *The time- t price of firm i dividend strip with maturity T , or equivalently time to maturity $\tau \equiv T - t$, is*

$$S_{i,t}(\tau) = D_{i,t} e^{w_0^i(\tau) + w_{\hat{x}}(\tau)\hat{x}_t + w_{\hat{x}_i}(\tau)\hat{x}_{i,t}}, \quad i \in \{V, G\}$$

with

$$w_{\hat{x}}(\tau) = \frac{1}{\lambda_x}(1 - e^{-\lambda_x \tau})(1 - 1/\psi), \quad w_{\hat{x}_i}(\tau) = \frac{1}{\lambda} \left(1 - e^{-\bar{\lambda} \tau}\right),$$

and $w_0^i(\tau)$ as defined in Appendix A.6. The corresponding dividend strip risk premium is equal to⁵

$$RP_i(\tau) = (\sigma_D + w_{\hat{x}}(\tau)\hat{\sigma}_x + w_{\hat{x}_i}(\tau)\hat{\sigma}_{x_i}) \Lambda.$$

Proof. See Appendix A.6.

The risk premium of firm i dividend strip is the sum of three parts. The term $\sigma_D \Lambda$ represents the reward for cashflow risk, a component that remains constant regardless of the maturity. The second part, $w_{\hat{x}}(\tau)\hat{\sigma}_x \Lambda$, accounts for the compensation investors receive for market-wide expected growth shocks. The third component, $w_{\hat{x}_i}(\tau)\hat{\sigma}_{x_i} \Lambda$, varies among firms. This variation stems from the heterogeneity in conditional covariances between the changes in the perceptions of a firm's expected growth and the cashflow growth surprises, as captured by $\hat{\sigma}_{x_i}$.

Corollary 3. *For every time to maturity $\tau > 0$, the slope of the equity term structure for*

⁵Note that the “consumption strip” price-to-consumption ratio is exponential-affine in the expected output growth component $\hat{x}$ only. Because the risk-free rate is affine in $\hat{x}$ only, the shape of the term structure of consumption strip risk premium is entirely driven by the dynamics of the risk-free rate.

firm i , defined as

$$RP_i(\tau) - \lim_{\tau \rightarrow 0} RP_i(\tau)$$

is decreasing in p_i , as long as the market price of risk for consumption growth shocks is positive ($\Lambda > 0$).

This corollary is a direct consequence of Corollary 1, which demonstrated that a lower p_i is linked with a higher $\hat{\sigma}_{x_i}$. When the signal is less informative, investors perceive the firm-specific expected growth component as being largely driven by priced aggregate shocks. It follows that, with all else being equal, the term structure of equity for a firm with lower signal precision will be higher than that for a firm with higher signal precision. Additionally, as the factor $w_{\hat{x}_i}(\tau)$ increases with τ , the difference between the two term structures will widen with the time horizon.

Proposition 5 provides the firms' stock prices, which can also be approximated by an exponential-affine function of the state variables, and the firms' risk premiums.

Proposition 5. *The time- t stock price of firm i is given by*

$$P_{i,t} = D_{i,t} e^{A_0^i + A_{\hat{x}}^i \hat{x}_t + A_{\hat{x}_i}^i \hat{x}_{i,t}}, \quad i \in \{V, G\}$$

with

$$A_{\hat{x}}^i = \frac{1 - 1/\psi}{1 - k_{1i}(1 - \lambda_x)}, \quad A_{\hat{x}_i}^i = \frac{1}{1 - k_{1i}(1 - \bar{\lambda})},$$

and A_0^i , k_{1i} as defined in Appendix A.8. The corresponding stock risk premium is

$$RP_i = (\sigma_D + A_{\hat{x}}^i \hat{\sigma}_x + A_{\hat{x}_i}^i \hat{\sigma}_{x_i}) \Lambda. \quad (9)$$

Proof. See Appendix A.8.

In line with Proposition 4, Proposition 5 shows that the first component of the risk premium corresponds to the cashflow growth risk, while the second and third components correspond to the risks associated with expected cashflow growth. When ψ is larger than one, the substitution effect dominates. An increase in expected cashflow growth due to an increase in $\hat{x}$ signals improved investment opportunities, which pushes the investor to invest more in risky assets. As a result, the price of the strip and the price of the stock respond positively to an increase in $\hat{x}$. This translates into a positive second risk premium component when $\psi > 1$. When ψ is smaller than one, the income effect dominates. An increase in expected

cashflow growth due to an increase in $\hat{x}$ signals higher future consumption. Because of her consumption smoothing motives, the investor increases current consumption and therefore decreases investments in risky assets. As a result, the price of the strip and the price of the stock respond negatively to an increase in $\hat{x}$. The third component of the risk premium is firm-specific and positive because the process $\hat{x}_i$ drives firm i 's expected cashflow growth.

The precision of the news signal, p_i , plays a significant role in determining the risk premium on firm i 's stock. Given that the return on the stock is a weighted average of the returns on dividend strips across all maturities, the logic of p_i influencing the risk premium on the dividend strips extends similarly to the overall stock risk premium. Although the risk premium formula, as specified in Equation (9), depends on p_i through the log-linearization constant k_{1i} as well, this influence is minimal. Therefore, our discussion will primarily focus on how p_i impacts the risk premium through $\hat{\sigma}_{x_i}$. The significance of this component hinges solely on the degree to which shocks to firm-specific expected growth are perceived by the agent to move in tandem with priced shocks. If there is no cashflow shock correlation ($\rho_i = 0$) and the signal is perfect, this third component does not add to the risk premium because $\hat{\sigma}_{x_i} = 0$. However, when the signal is imprecise, differentiating between priced and unpriced shocks becomes challenging. As a result, the firm-specific risk affects the risk premium. Its contribution is most significant when the signal precision p_i approaches zero, because $\hat{\sigma}_{x_i}$ is inversely related to p_i , as shown in Corollary 1.

3 Data

We use data gathered from several sources: GDP and GDP forecast data is from the Survey of Professional Forecasters. Information on firm earnings and analyst forecasts is obtained from the Institutional Brokers' Estimate System (IBES). Data on firm characteristics and stock prices are from the Center for Research in Securities Prices (CRSP) and Compustat. The Consumer Price Index data, which allows conversion to real values, is from Robert Shiller's website.

We require that firm's common shares, as indicated by CRSP share codes 10 and 11, are listed on the New York Stock Exchange (NYSE), American Stock Exchange, or NASDAQ. Additionally, sufficient accounting data must be available to categorize the stock into one of the three book-to-market portfolios: the bottom 30%, the middle 40%, and the top 30%.

3.1 Book-to-Market Data

Following the [Fama and French \(1993\)](#) methodology, companies are categorized annually, in June, based on their book-to-market ratios. This ratio compares a company’s book equity at the end of its last fiscal year with its market equity at the end of the previous December.

Market equity is obtained from CRSP data. To calculate book equity from Compustat data, we start with the company’s stockholders’ equity and adjust it by adding any deferred taxes and investment tax credits, then subtract the book value of preferred stock. If direct equity data is not available, we estimate it using the sum of common equity and preferred stock’s par value, or by subtracting total liabilities from total assets. This approach follows the methods used by [Davis, Fama, and French \(2000\)](#) and [Akey, Robertson, and Simutin \(2024\)](#).⁶

To categorize the companies, we use the NYSE breakpoints. Specifically, the book-to-market ratio thresholds are determined by the 30th and 70th percentiles of the book-to-market ratios for NYSE-listed stocks. Consequently, all companies are classified into three groups based on their book-to-market ratios: low (bottom 30%), medium (middle 40%), and high (top 30%).

3.2 Earnings and Forecast Data

Our dataset on company earnings and forecasts spans from the first quarter of 1984 to the fourth quarter of 2022, with a focus on quarterly information provided by IBES unadjusted files. This includes actual earnings data from the Unadjusted Detail Actual files and analyst earnings forecasts from the Unadjusted Detail History files.⁷

We retrieve the latest analyst forecast for each firm, focusing on quarterly forecasts for both the current and the upcoming fiscal quarters, indicated by ‘fpi’ codes 6 and 7. To ensure consistency in our analysis, we determine a primary reporting basis for each security, categorizing forecasts as either based on primary (P) or diluted (D) shares.

To compare forecasts with actual earnings accurately, we merge the actuals and forecasts files and then adjust the forecast figures for per-share consistency.⁸ This step involves using CRSP’s cumulative adjustment factors, accounting for stock splits, to align forecasts with actual company-reported earnings per share. We also adjust forecast and earnings

⁶For more details, see: <http://www.law.uchicago.edu/law-finance/code/NoisyFactors>

⁷We rely on IBES data on earnings for our analysis, as it provides a more extensive and detailed dataset compared to the information available on dividends. Using the Unadjusted Detail History and Actuals in conjunction with the CRSP cumulative adjustment split factor, sourced from the CRSP Daily files, ensures the highest level of reliability and accuracy in our analysis.

⁸Details on this step are available at [WRDS website](https://wrds-www.wharton.upenn.edu/documents/5/A_Note_on_IBES_Unadjusted_Data_pdf.pdf). See also documentation on IBES available at https://wrds-www.wharton.upenn.edu/documents/5/A_Note_on_IBES_Unadjusted_Data_pdf.pdf

announcement dates to the nearest trading day in CRSP’s daily data, ensuring no missing share adjustment factors.

We define the consensus forecast as the median of all forecasts made by analysts within the 90 days leading up to the earnings announcement. For analysts who revise their estimates in this timeframe, only their final revision is considered.

For the analysis across book-to-market categories, we multiply the per-share earnings and consensus forecast by the total outstanding shares at the time of the earnings report using CRSP data. These figures are then aggregated to construct quarterly time series for each book-to-market category. That is, the earnings and earnings forecast of the value portfolio are defined as respectively the total earnings and total earnings forecast of the high (top 30%) book-to-market ratio category. Similarly, the earnings and earnings forecast of the growth portfolio are defined as respectively the total earnings and total earnings forecast of the low (bottom 30%) book-to-market ratio category. Summary statistics on the growth forecast error and growth forecast of GDP, value portfolio earnings, and growth portfolio earnings are reported in Appendix B.

3.3 News Data

To quantify the number of information signals received by investors and to provide additional empirical support for our model, we use company-related news data from RavenPack. The data spans from 2000 to 2022 and includes coverage from sources such as Dow Jones Newswires and The Wall Street Journal. The data records daily media mentions, which we aggregate to a quarterly frequency to align with the periodicity of earnings and forecast data. Types of news coverage encompass hot news flashes (urgent news stories labeled as “breaking news” during the editorial process), news flashes, which are headlines without accompanying body text, and full articles.

4 Results

In this section, the parameters of the model are estimated by maximum likelihood. The estimated parameter values yield model-implied economic and financial moments that are close to their empirical counterparts. The model predicts a market risk premium of about 7%, a value premium of about 4%, an upward-sloping term structure of equity risk premium for value firms, and a flat term structure of equity risk premium for growth firms.

Table 1: Estimated Parameter Values.

This table reports the parameter values estimated by maximum likelihood using GDP and GDP forecasts, earnings and earnings forecasts of the value portfolio, and earnings and earnings forecasts of the growth portfolio. The data is in real terms and at the quarterly frequency from Q1:1984 to Q4:2022. Standard errors are reported in parentheses. All parameters are statistically significant at the 1% level.

Parameter	Symbol	Value
Output growth volatility	σ_C	0.0199 (0.0003)
Mean output growth	μ	0.0260 (0.0030)
Expected output growth volatility	σ_x	0.0124 (0.0003)
Expected output growth reversion speed	λ_x	0.2320 (0.0160)
Cashflow growth volatility	σ_D	0.1659 (0.0042)
Expected cashflow growth volatility	$\bar{\sigma}$	0.1270 (0.0015)
Expected cashflow growth reversion speed	$\bar{\lambda}$	0.3309 (0.0107)
Firm V signal precision	p_V	0.5100 (0.0058)
Firm G signal precision	p_G	0.8977 (0.0212)
Firm V cashflow shock correlation	ρ_V	-0.2623 (0.0293)
Firm G cashflow shock correlation	ρ_G	-0.6521 (0.0231)

4.1 Parameter Estimation

The parameters of the model are estimated by maximum likelihood using U.S. GDP, GDP forecast, earnings, and earnings forecast data. The GDP growth rate and forecasted GDP growth rate are used to build proxies for the output growth rate, $\log(C_{t+\Delta}/C_t)$, and expected output growth component, $\hat{x}_t\Delta$, in the model. The earnings growth rate and forecasted earnings growth rate of the growth (resp., value) portfolio are used to build proxies for the growth firm realized cashflow growth, $\log(D_{G,t+\Delta}/D_{G,t})$, (resp., value firm realized cashflow growth, $\log(D_{V,t+\Delta}/D_{V,t})$) and growth-firm-specific expected growth component, $\hat{x}_{G,t}\Delta$, (resp., value-firm-specific expected growth component, $\hat{x}_{V,t}\Delta$) in the model. The data is in real terms and at the quarterly frequency from Q1:1984 to Q4:2022, implying that $\Delta = 1/4$. Additional

details on the estimation are provided in Appendix C.

Table 1 reports the estimated parameter values. All parameters are statistically significant at the 1% level. The mean output growth is about 2.5%, the output growth volatility is close to 2%, and firms' cashflow growth volatility is approximately 16%. The estimated mean-reversion speeds show that the expected growth components $\hat{x}$, $\hat{x}_V$, and $\hat{x}_G$ are persistent, as in the long-run risk literature. The signal precision of the value firm is substantially lower than the signal precision of the growth firm. The value firm's cashflow shock correlation is negative but less so than the growth firm's cashflow shock correlation. As detailed in the next sections, these features imply that the term structure of equity risk premium is flatter for the growth firm than for the value firm, and that the difference between the value firm's risk premium and the growth firm's risk premium—the value premium—is about 4%.

The preference parameters are set in line with the existing literature. The relative risk aversion is $\gamma = 6$, the elasticity of intertemporal substitution is $\psi = 1.1$, and the subjective discount factor is $\delta = 0.97$. As shown in what follows, these preference parameters together with the estimated parameter values provided in Table 1 yield model-implied levels for the market risk premium and value premium that are reasonably close to their empirical counterparts.

Figure 1 depicts the term structure of output growth variance ratios obtained in the model (solid line) and in the data using the GDP (dashed line).⁹ The model-implied term structure of output growth variance ratios is upward-sloping, meaning that output growth risk is particularly high in the long term relative to the short term. The reason is that the estimated mean-reversion speed λ_x is relatively low (see Table 1), yielding a persistent expected output growth component $\hat{x}$. As a result, output growth risk is concentrated in the long term, as in the long-run risk literature. The model-implied term structure of output growth variance ratios aligns closely with its empirical counterpart. This shows that the model captures reasonably well the timing of output growth risk observed in the data.

Table 2 provides output growth, market cashflow growth, and market return moments both in the model and in the data. In the model, output growth mean is 2.6%, output growth volatility is 2.2%, market cashflow growth mean is 3.4%, and market cashflow growth volatility is about 17.3%. These model-implied moments are reasonably close to their empirical

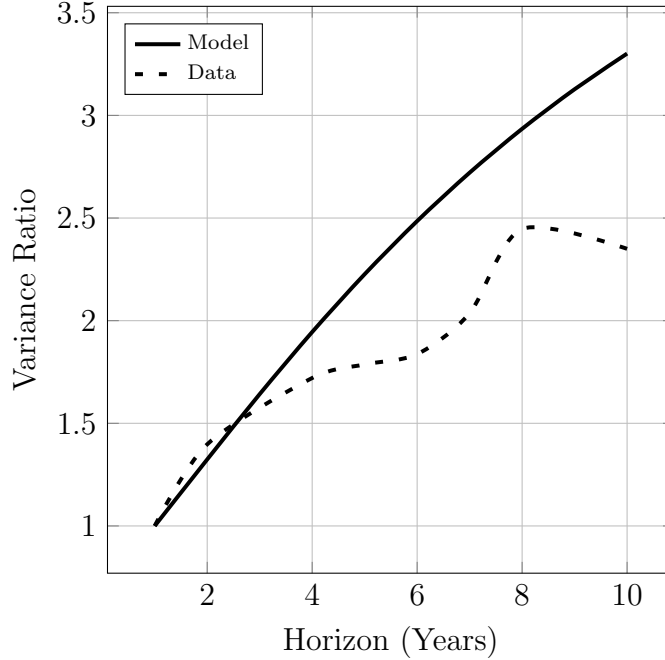
⁹In the model, variance ratios are defined as

$$\widehat{VR}_C(t, \tau) = \frac{\hat{\sigma}_C^2(t, \tau)}{\hat{\sigma}_C^2(t, 1)},$$

where, as in Belo et al. (2015) and Marf  (2016) among others, the variances $\hat{\sigma}_C(t, \tau) = \sqrt{\frac{1}{\tau} \log \left(\frac{\mathbb{E}[C_{t+\tau}^2 | \mathcal{F}_t^C]}{\mathbb{E}[C_{t+\tau}^2 | \mathcal{F}_t^C]^2} \right)}$ are computed via the moment-generating function of C .

Figure 1: Term Structure of Output Growth Variance Ratios.

This figure depicts the term structure of output growth variance ratio in the model (solid line) and in the data using the GDP (dashed line). As for the estimation, GDP data is in real terms and at the quarterly frequency from Q1:1984 to Q4:2022. The model parameters are provided in Section 4.1.



counterparts provided in the first column of the table. This shows that the model generates realistic output growth and market cashflow growth moments.

The dynamics of the firm-specific expected growth components, $\hat{x}_V$ and $\hat{x}_G$, provided in Equation (8) together with the estimated parameter values reported in Table 1 imply that the exposure of $\hat{x}_V$ to the expected output growth component $\hat{x}$ is higher than that of $\hat{x}_G$. In other words, the correlation between the value firm expected growth component $\hat{x}_V$ and the expected output growth component $\hat{x}$ is larger than the correlation between the growth firm expected growth component $\hat{x}_G$ and the expected output growth component $\hat{x}$. The reason is that the value firm news signal precision is lower than that of the growth firm. To test this prediction of the model, we compare the empirical and model-implied correlation between $\hat{x}_V$ (resp., $\hat{x}_G$) and $\hat{x}$. The correlation between $\hat{x}_V$ and $\hat{x}$ is 0.41 in the data and 0.68 in the model, whereas the correlation between $\hat{x}_G$ and $\hat{x}$ is 0.03 in the data and 0.07 in the model. This shows that the maximum likelihood estimation generates model-implied correlations between $\hat{x}$, $\hat{x}_G$, and $\hat{x}_V$ that are reasonably close to their empirical counterparts. Importantly, the exposure of the value firm expected growth component to the expected output growth component is substantially larger than that of the growth firm expected

Table 2: Moments of Cashflow Growth and Cashflow Growth Forecast.

In the data, output is GDP, market cashflow is the total earnings of firms in CRSP–IBES–Compustat, firm G cashflow is the earnings of the growth portfolio, and firm V cashflow is the earnings of the value portfolio (see Section 3.2). $\hat{x}$ is the expected output growth component, $\hat{x}_G$ is the growth-firm-specific expected growth component, and $\hat{x}_V$ is the value-firm-specific expected growth component. As for the estimation, GDP and earnings data is in real terms and at the quarterly frequency from Q1:1984 to Q4:2022. The first column provides the empirical moments, whereas the second column provides the model-implied counterparts. The model parameters are provided in Section 4.1. Moments are annualized.

	Data	Model
Output growth mean	0.0255	0.0260
Output growth volatility	0.0218	0.0217
Market cashflow growth mean	0.0504	0.0336
Market cashflow growth volatility	0.1835	0.1733
$corr(\hat{x}, \hat{x}_G)$	0.0282	0.0732
$corr(\hat{x}, \hat{x}_V)$	0.4074	0.6812
$vol(\hat{x}_G)$	0.1261	0.1065
$vol(\hat{x}_V)$	0.1234	0.1049
$vol(\hat{x})$	0.0121	0.0126

growth component in both the model and the data.

In addition, because the signal precision of the growth firm is larger than that of the value firm, the agent learns better about the growth firm unobservable component x_G than about the value firm unobservable component x_V . That is, the agent updates her belief about x_G more aggressively than about x_V , which means that $\hat{x}_G$ is more volatile than $\hat{x}_V$ in the model (Corollary 2). The volatility of $\hat{x}_G$ is 12.61% in the data and 10.65% in the model, whereas the volatility of $\hat{x}_V$ is 12.34% in the data and 10.49% in the model. Consistent with the prediction of the model, the empirical volatility of $\hat{x}_G$ is slightly larger than that of $\hat{x}_V$. The volatility of expected output growth $\hat{x}$ is 1.21% in the data and 1.26% in the model. That is, the maximum likelihood estimation generates a model-implied expected output growth volatility that is reasonably close to its empirical counterpart.

To summarize, the firm-specific expected growth components $\hat{x}_V$ and $\hat{x}_G$ extracted from

analyst earnings forecasts have statistical properties that are in line with the learning model.

4.2 Value Premium and Equity Term Structures

This section discusses the value premium and the equity term structures of value firms, growth firms, and the market. The model generates a value premium of about 4%, consistent with empirical estimates. The model-implied equity term structure is upward-sloping for value firms and flat for growth firms.

4.2.1 Different Signal Precisions Only

The first goal is to highlight how information processing and learning with different signal precisions determine the valuation of a firm and the level of its risk premium.¹⁰ Figure 2 depicts the dividend yield (i.e., cashflow-to-price ratio) and risk premium of firm i as a function of the news signal precision p_i . The news signal precision p_i is the only source of cross-sectional heterogeneity throughout this section; the cashflow shock correlation is constant and set to $\rho_i = 0$. Both the dividend yield and risk premium decrease with the signal precision. Thus, we see that learning endogenously gives rise to the classification of firms into value (high fundamental to price ratio) and growth (low fundamental to price ratio) categories. Firms featuring low news signal precision have high dividend yields and are therefore interpreted as value firms. Conversely, firms featuring high news signal precision have low dividend yields and are therefore interpreted as growth firms.¹¹

The relationship between signal informativeness, valuations, and risk premium depicted on Figure 2 arises due to imperfect information and learning. The informativeness of the signal affects the agent's perception of priced versus unpriced shocks. Intuitively, a highly informative signal (p_i close to 1) on the firm-specific expected growth shocks implies that the agent is close to perfectly observing these shocks. As a result, the agent does not require a large risk premium to hold the firm's stock and bear these shocks; the firm's valuations are relatively high. Conversely, if the precision of the signal is low (p_i close to 0), the agent is not able to discern firm-specific expected growth shocks. Therefore, the agent requires a large risk premium and low valuations to hold the firm's stock.

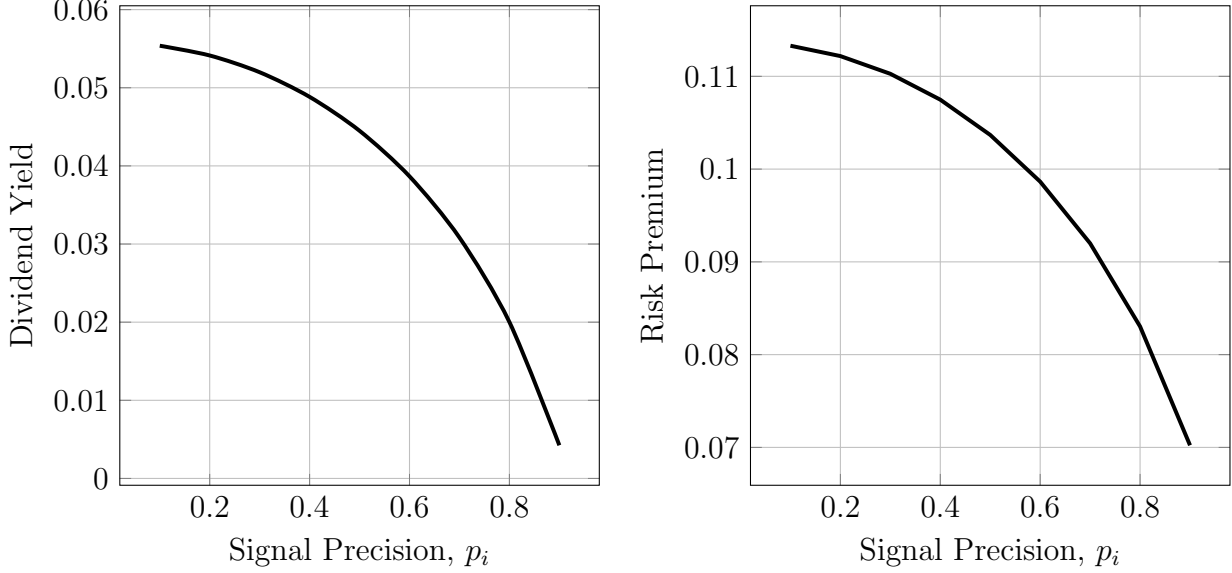
To further understand the economic mechanism, let us consider two firms: V and G . Suppose that firm V features a completely uninformative signal ($p_V = 0$), whereas firm G

¹⁰Complementary to our study, Kiku (2007) shows that considering heterogeneous cashflow parameters across firms can explain observed cross-sectional differences in firms' return volatilities, return covariances, and CAPM-betas.

¹¹In accordance with the literature (see, e.g., Lettau and Wachter, 2007, 2011; Tsai and Wachter, 2015), the dividend yield is the fundamental-to-price metric that defines the value and growth categories in our equilibrium model.

Figure 2: Cross-Section of Dividend Yield and Risk Premium.

This figure depicts the impact of the signal precision on the model-implied dividend yield (left panel), and risk premium (right panel). Firm i cashflow shock correlation is $\rho_i = 0$. The other parameters are provided in Section 4.1.



features a perfectly informative signal ($p_G = 1$). These two extreme signal precisions imply that the correlation between the realized cashflow growth rate and the firm-specific expected cashflow growth component is very different across firms (a consequence of Corollary 1). Since firm V 's news signal is uninformative, the agent updates her beliefs about $\hat{x}_V$ using a single piece of information, namely the cashflow growth surprise $d\hat{B}$. As a result, the expected cashflow growth component $\hat{x}_V$ is perfectly correlated to firm V 's realized cashflow growth rate (see Proposition 2). In contrast, the agent updates her beliefs about $\hat{x}_G$ by relying heavily on the news signal. Therefore, the expected cashflow growth component $\hat{x}_G$ is uncorrelated to firm G 's realized cashflow growth rate (see Proposition 2).

To investigate the impact of these different correlations on the dividend yields and risk premiums of firms V and G , let us consider a negative cashflow growth surprise $d\hat{B}$. This negative surprise implies that firm V 's expected growth component $\hat{x}_V$ is updated downward, whereas firm G 's expected growth component $\hat{x}_G$ is unaffected. That is, the negative surprise impacts strongly and negatively the expected cashflow growth component of firm V , but does not impact the expected cashflow growth component of firm G . As a result, the agent requires a higher risk premium and lower valuations (i.e., higher dividend yield) to hold firm V 's stock than to hold firm G 's stock.

To quantify the impact of learning with different signal precisions on the pricing of firms

V and G , the signal precisions are now set to their estimated values reported in Table 1: $p_V = 0.5100$ and $p_G = 0.8977$. Again, the cashflow shock correlations of firms V and G are assumed to be equal: $\rho_V = \rho_G = 0$. The model-implied dividend yield and risk premium of firm V are 4.40% and 10.32%, respectively. The dividend yield of firm G is 0.47%, and its risk premium is 7.06%. In this model of learning with different signal precisions, firms V and G endogenously emerge as respectively value and growth firms. The model-implied risk premiums of the value and growth firms yield a model-implied value premium of 3.26%. In the data, the value premium, measured as the average return of the HML portfolio (Fama and French, 1993), is 4.12%. This shows that the model of learning with different signal precisions only ($p_V \neq p_G$ and $\rho_V = \rho_G = 0$) can explain about 80% of the value premium observed in the data. In Section 4.2.2 below, we show that learning with different signal precisions and cashflow shock correlations ($p_V \neq p_G$ and $\rho_V \neq \rho_G$) increases the model-implied value premium to about 4%, in line with the data. When $\rho_V = \rho_G = 0$, the model-implied market risk premium is 7.37%.¹² In the data, the market risk premium, measured by the average excess return of the CRSP market index, is 8.05%.

The top left panel of Figure 3 depicts the equity term structures for the value firm V (solid line), growth firm G (dashed line), and market (dotted line) when the cashflow shock correlations are equal: $\rho_V = \rho_G = 0$. The equity term structure of the value firm is markedly upward-sloping, that of the growth firm is moderately upward-sloping, and that of the market is by definition a value-weighted average of the equity term structures of the value and growth firms.¹³ The bottom panel of Figure 3 shows that the firms' equity term structure slope decreases with the signal precision. That is, an increase in signal precision implies a flatter equity term structure (Corollary 3).

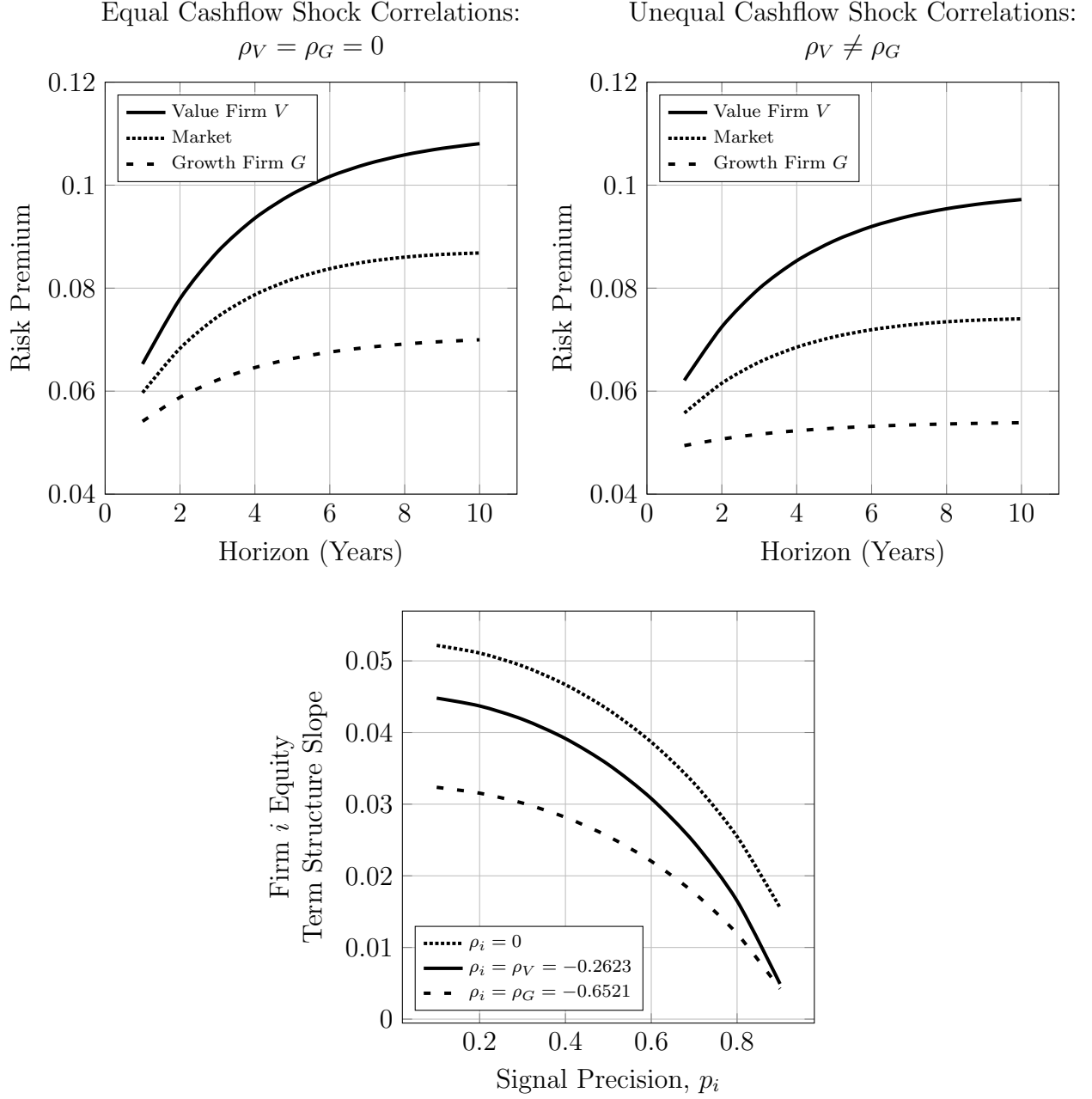
The economic mechanism is as follows. When the news signal about x_i is close to being uninformative (low signal precision p_i), the agent updates her filter $\hat{x}_i$ using mainly informa-

¹² Because the market dividend is the sum of firm V dividend and firm G dividend, $D_{m,t} = D_{V,t} + D_{G,t}$, the stock market price is the sum of the two firms' stock prices: $P_{m,t} = P_{V,t} + P_{G,t}$, where $P_{V,t}$ and $P_{G,t}$ are provided in Proposition 5. Consequently, the risk premium of the stock market is the following *value-weighted* average of the risk premiums of firm V stock and firm G stock: $RP_{m,t} = \frac{P_{V,t}}{P_{m,t}} RP_V + \frac{P_{G,t}}{P_{m,t}} RP_G$, where RP_V and RP_G are defined in Proposition 5. Note that firm V dividend and firm G dividend determining the stock prices $P_{V,t}$ and $P_{G,t}$, respectively, are assumed to be equal: $D_{V,t} = D_{G,t} = 1$. In addition, the expected growth components are set to their long-term mean: $\hat{x}_t = \hat{x}_{V,t} = \hat{x}_{G,t} = 0$.

¹³ Because the market dividend is the sum of firm V dividend and firm G dividend, $D_{m,t} = D_{V,t} + D_{G,t}$, the price of a market dividend strip at time t with time to maturity τ is the sum of the two firms' dividend strips: $S_{m,t}(\tau) = S_{V,t}(\tau) + S_{G,t}(\tau)$, where $S_{V,t}(\tau)$ and $S_{G,t}(\tau)$ are provided in Proposition 4. Consequently, the risk premium of the market dividend strip is the following *value-weighted* average of the risk premiums of firm V dividend strip and firm G dividend strip: $RP_{m,t}(\tau) = \frac{S_{V,t}(\tau)}{S_{m,t}(\tau)} RP_V(\tau) + \frac{S_{G,t}(\tau)}{S_{m,t}(\tau)} RP_G(\tau)$, where $RP_V(\tau)$ and $RP_G(\tau)$ are defined in Proposition 4. Note that firm V dividend and firm G dividend determining the dividend strip prices $S_{V,t}(\tau)$ and $S_{G,t}(\tau)$, respectively, are assumed to be equal: $D_{V,t} = D_{G,t} = 1$. In addition, the expected growth components are set to their long-term mean: $\hat{x}_t = \hat{x}_{V,t} = \hat{x}_{G,t} = 0$.

Figure 3: Equity Term Structures.

The top panels depict the model-implied term structures of risk premium for the value firm V (solid line), growth firm G (dashed line), and market (dotted line). The signal precision is assumed to be the only source of heterogeneity across the two firms in the top left panel; cashflow shock correlations are equal: $\rho_V = \rho_G = 0$. Both the signal precisions and the cashflow shock correlations are different across the two firms in the top right panel; the parameters are provided in Section 4.1. The bottom panel depicts the slope of the equity term structure of firm i , defined as the difference between the 10-year dividend strip risk premium and the 1-year dividend strip risk premium of firm i , as a function of the signal precision p_i . The dotted, solid, and dashed lines are obtained with the cashflow shock correlations $\rho_i = 0$, $\rho_i = \rho_V = -0.2623$, and $\rho_i = \rho_G = -0.6521$, respectively.



tion provided by the cashflow growth shock $d\widehat{B}$ (see Proposition 2 and Corollary 1). That is, the filter $\widehat{x}_i$ is highly correlated to the output shock, which is a priced source of risk. Therefore, the risk of $\widehat{x}_i$ contributes to the firm's risk premium when the signal precision is low (see Proposition 4). As a negative shock hits the firm's current cashflow, the expected future cashflow growth components $\widehat{x}$ and $\widehat{x}_i$ are revised downward. This means that the current negative shock impacts also future cashflows negatively and strongly, via the impact of both $\widehat{x}$ and $\widehat{x}_i$. As a result, cashflow risk is substantially larger in the long run than in the short run for the value firm V . Because shocks in both $\widehat{x}$ and $\widehat{x}_i$ are priced, the substantially larger cashflow risk in the long run than in the short run translates into a markedly upward-sloping term structure of risk premium. In contrast, when the news signal about x_i is highly informative (high signal precision p_i), the agent updates her filter $\widehat{x}_i$ using mainly information provided by the signal (see Proposition 2 and Corollary 1). Since the news signal is uncorrelated to the output shock, signal shocks are not priced in equilibrium. Therefore, the firm's risk premium obtained with a highly informative signal depends only weakly on the risk of the filter $\widehat{x}_i$ (see Proposition 4). It, however, still depends on the risk of the filter $\widehat{x}$, which, as detailed above, has an upward-sloping effect. This explains why the growth firm G has a term structure of risk premium that is upward-sloping as well, but less markedly so than that of the value firm V .

4.2.2 Different Signal Precisions and Cashflow Shock Correlations

This section discusses the value premium and the equity term structures obtained with both different signal precisions and different cashflow shock correlations. The parameters of the model are provided in Section 4.1. That is, the assumption that the signal precision is the only source of heterogeneity across the two firms is relaxed.

The model-implied dividend yield and risk premium of the value firm V are 4.76% and 9.28%, respectively. The model-implied dividend yield of the growth firm G is 2.83%, and its risk premium is 5.37%. The risk premiums of firms V and G yield a model-implied value premium of 3.91%, which is close to the value premium of 4.12% observed in the data. In addition, the market risk premium of 6.82% in the model and 8.05% in the data.

The right panel of Figure 3 depicts the equity term structures for the value firm V (solid line), growth firm G (dashed line), and market (dotted line) when both the signal precisions and cashflow shock correlations are different across the two firms. The equity term structure of the value firm is markedly upward-sloping for the reasons detailed in Section 4.2.1. In contrast, the equity term structure of the growth firm is flat.

The increase in the value premium from 3.26% in Section 4.2.1 with cashflow shock correlations $\rho_V = \rho_G = 0$ to 3.91% in the current section with $\rho_V = -0.2623$ and $\rho_G =$

-0.6521 is explained as follows. The estimated correlation for the value firm V is slightly negative, whereas that of the growth firm G is substantially negative. This indicates that a negative shock in realized cashflow growth tends to generate positive growth in the future, particularly so for the growth firm. If investors can discern this tendency—more feasible for the growth firm than for the value firm due to the high precision of the news signal p_G —there is a strong incentive to invest in this firm’s stock. This results in a substantially lower risk premium for the growth firm and a slightly lower risk premium for the value firm, thereby explaining the higher value premium obtained with heterogeneous cashflow shock correlations.

Figure 3 shows that the equity term structure of the value firm V flattens out but remains markedly upward-sloping, as its cashflow shock correlation changes from 0 to -0.2623 . The substantial negative cashflow shock correlation of the growth firm, coupled with its high signal precision, yields a flat equity term structure. The flattening effect comes from the fact that the agent learns with high precision that bad cashflow shocks today are likely to be followed by good shocks tomorrow. This means that for the growth firm G , long-term cashflow growth rates have about the same risk as short-term ones, which translates into similar risk premiums in the long term and in the short term.

Related to our study, Giglio et al. (2024) find that the equity term structure of the HML portfolio—long value firms and short growth firms—increases by a magnitude of about 0.8 percentage point between the 5-year and the 10-year maturities, and by about 1.8 percentage points between the 5-year and the 15-year maturities.¹⁴ In comparison, the equity term structure of the HML portfolio in our model—long firm V and short firm G —increases by 0.70 percentage points between the 5-year and the 10-year maturities, and by 0.83 percentage points between the 5-year and the 15-year maturities.

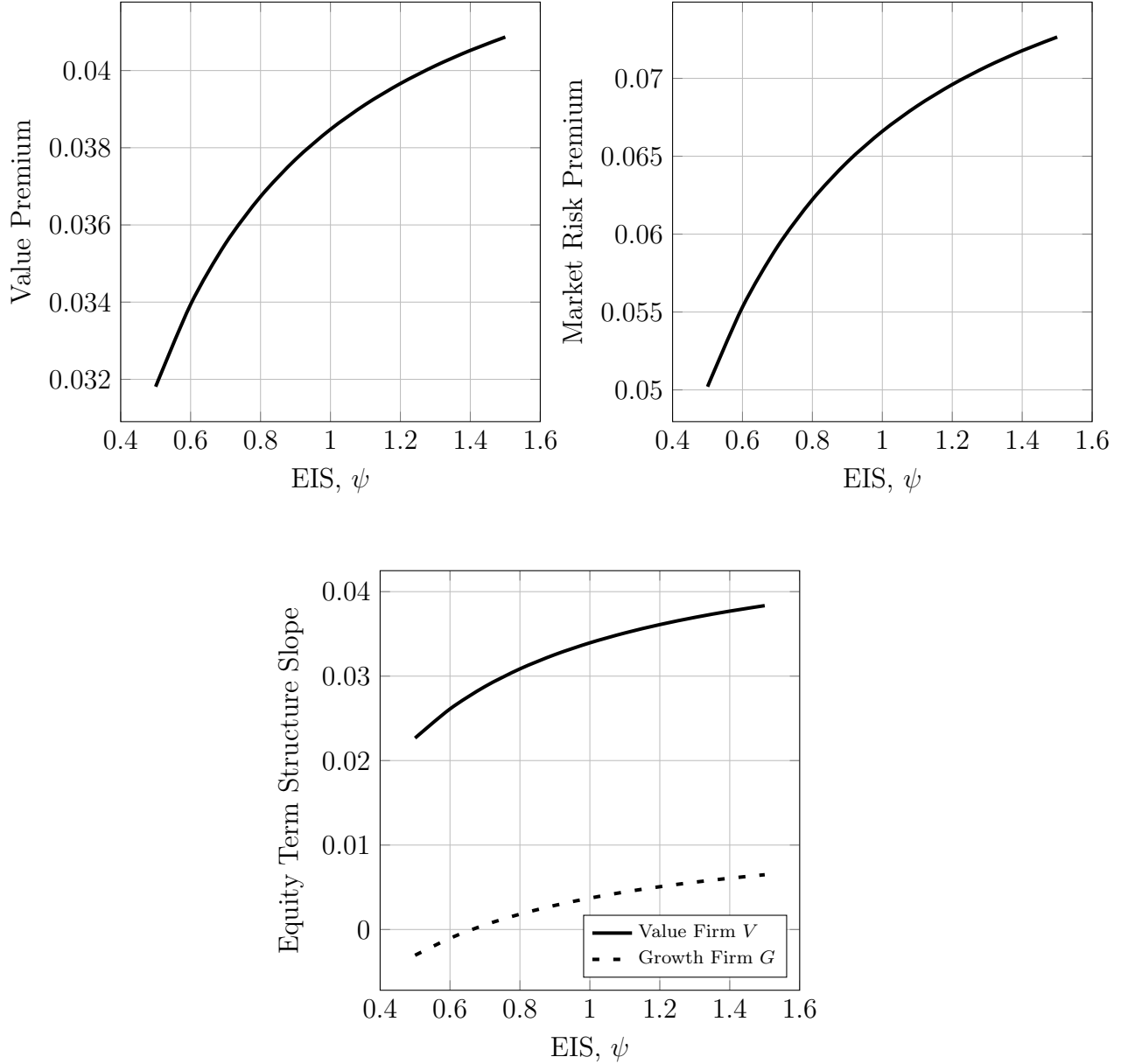
The model-implied equity term structure of the market is a weighted-average of the equity term structures of the value and growth firms. As a result, the equity term structure of the market is upward-sloping in the upper-right panel of Figure 3. Whether the observed equity term structure of the market is upward-sloping or not is the subject of a debate (see, e.g., van Binsbergen et al., 2012, 2013; van Binsbergen and Koijen, 2017; Gormsen, 2021; Bansal et al., 2021; Boguth et al., 2023). Recently, Giglio et al. (2024) find that the equity term structure of the market increases monotonically between the 1-year and 15-year maturities,¹⁵ similar to the prediction of our model.

¹⁴It is worth noting that while the point estimates of the risk premiums increase between the 5-year and 15-year maturities, none of the risk premiums are statistically different from one another, and therefore the increase is not statistically significant.

¹⁵Here again, while the point estimates of the risk premiums increase with the maturity, none of the risk premiums are statistically different from one another, and therefore the increase is not statistically significant.

Figure 4: Results Sensitivity to the Elasticity of Intertemporal Substitution.

The model-implied value premium, market risk premium, and slope of the equity term structure are depicted as a function of the elasticity of intertemporal substitution (EIS), ψ . The other parameters are provided in Section 4.1. The slope of the equity term structure in the bottom panel is defined as the difference between the 10-year dividend strip risk premium and the 1-year dividend strip risk premium.



A related consideration is how firm size shapes the equity term structure. In the model, firm size is the product of the firm's dividend and the firm's price-dividend ratio, as defined in Proposition 5. When dividends are set to $D_V = D_G = 1$, firm G is larger than firm V because firm G 's price-dividend ratio is larger than that of firm V . Consequently, the small firm V has an upward-sloping equity term structure (Figure 3, right panel). This contrasts with Giglio et al. (2024), who find that small firms tend to have downward-sloping equity

term structures, though this decline in risk premiums across maturities is not statistically significant. In the model, the relative firm sizes are determined by the relative scaling of dividends. Increasing D_V relative to D_G such that $D_V \gg D_G$ would make firm V larger than firm G without affecting the firms price-dividend ratios or expected returns. In other words, when $D_V \gg D_G$, the small firm G has a flat equity term structure in our model (Figure 3, right panel).

Finally, Figure 4 depicts the sensitivity of the value premium, market risk premium, and slope of the equity term structure to the agent’s elasticity of intertemporal substitution (EIS), ψ . The other parameters of the model are provided in Section 4.1. As the EIS increases, the agent dislikes long-run cashflow risk more (Bansal and Yaron, 2004). Since long-run cashflow risk is particularly high for the value firm V , both its long-term dividend strip risk premium and stock risk premium increase markedly. This in turn implies an increase in the slope of the equity term structure of the value firm V and in the value premium. Since the growth firm G features moderate long-run cashflow risk, its equity term structure remains relatively flat as the EIS increases. Nevertheless, it is worth noting that the equity term structure of the growth firm G is moderately downward-sloping when the EIS is below 0.7, and moderately upward-sloping when the EIS is above 0.7. To conclude, for an EIS between 0.5 and 1.5, the value premium ranges from 3.2% to 4.1%, the market risk premium ranges from 5% to 7.3%, the equity term structure of the value firm V is markedly upward-sloping, and the equity term structure of the growth firm G is flat.

4.3 Consistent Evidence of Heterogeneity in Information Quality

The estimated signal precision for the high book-to-market (value) category, p_V , is substantially lower than that for the low book-to-market (growth), p_G (see Table 1). This section shows that this relationship between the quality of information and the firm’s characteristic is consistent with firm-level empirical evidence. Our analysis is supported by two pieces of evidence. First, we observe heterogeneity in news coverage and analyst forecast metrics across value and growth firms, as depicted in Figure 5. Second, firm-level regressions show different responses of analyst earnings forecasts for value versus growth firms to output shocks, as detailed in Table 3.

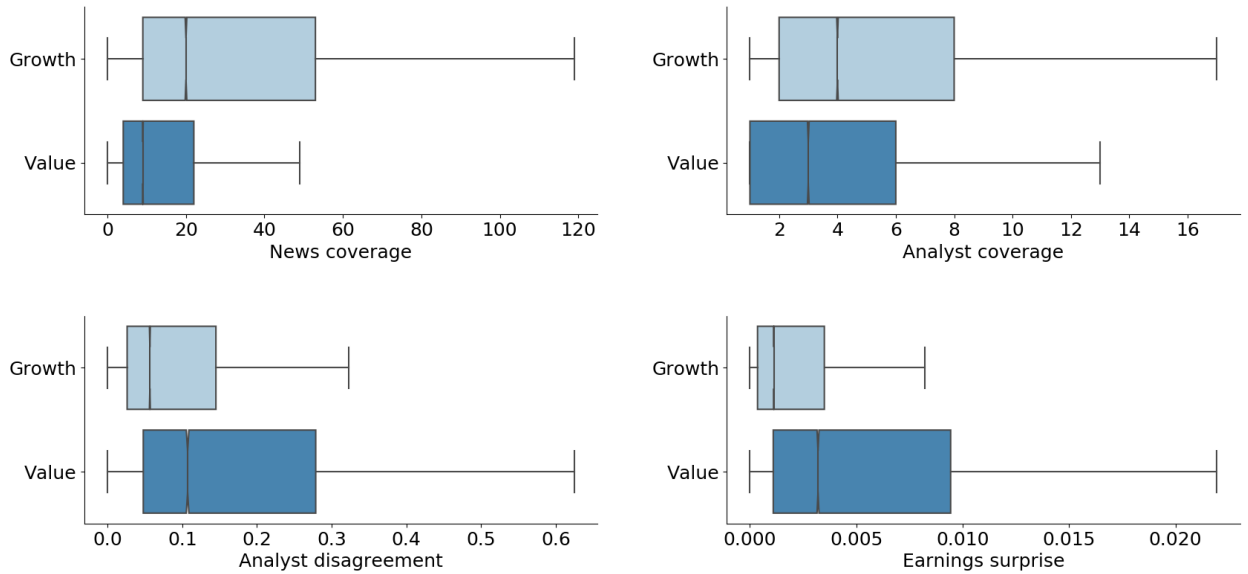
Turning to the first piece of evidence, Figure 5 illustrates the distributions of news coverage, analyst coverage, analyst disagreement, and earnings surprises across book-to-market categories.¹⁶ The data on news coverage is from RavenPack, aggregated quarterly from 2000 to 2022. Analyst forecast metrics are computed using the same quarterly earnings and fore-

¹⁶This figure shows the overall distribution across all years. The patterns remain stable over time when the data is divided into different time periods.

casts data from 1984 to 2022 that was used to construct the time series of earnings and earnings forecasts for the value and growth portfolios (Section 3).

Figure 5: News and Analyst Forecasts Metrics: Coverage, Disagreement, Earnings Surprises.

This figure plots the empirical distributions of news coverage, analyst coverage, analyst disagreement, and earnings surprises, across low and high book-to-market firms. All data is at a quarterly frequency, spanning from 2000 to 2022 for news coverage and from 1984 to 2022 for analyst forecasts. Analyst disagreement (dispersion) is calculated for firm-quarter observations with at least three analyst forecasts. In each box plot, the median is marked as a vertical line; the box extends to the quartiles of the data set, whereas the whiskers extend to an additional 1.5 times the inter-quartile range.



The upper panels of Figure 5 show that value firms generally receive fewer news and analyst coverage than growth firms. News coverage is defined as the cumulative count of news mentions for a particular firm within a given quarter. This includes mentions across news sources such as Dow Jones Newswires and The Wall Street Journal. Analyst coverage refers to the total number of analysts making earnings forecasts for a firm in a given quarter. We interpret both of these metrics as capturing the number of information signals received by investors.

Lower news and analyst coverage is consistent with lower precision of information investors have about value firms. To see why, assume that investors receive N_i signals s_i^j , for $i \in \{V, G\}$ and $j = 1, \dots, N_i$. Each signal s_i^j is the j -th noisy signal providing information on expected cashflows shocks of firm i . To simplify the exposition, let us assume that all

signals have equal precision, described by the following dynamics:

$$ds_{i,t}^j = \pi dB_{x_i,t} + \sqrt{1 - \pi^2} dB_{i,t}^j,$$

where $0 < \pi < 1$ represents the accuracy of each signal, and all Brownian motions are uncorrelated. Investors aggregate these N_i information sources into an aggregate signal $s_{i,t}$, which evolves according to:

$$ds_{i,t} = p_i dB_{x_i,t} + \sqrt{1 - p_i^2} dB_{s_i,t}$$

for $i \in \{V, G\}$.¹⁷ The precision of this aggregate signal, p_i , is given by:

$$p_i = \frac{\pi}{\sqrt{\frac{1}{N_i}(1 + (N_i - 1)\pi^2)}}. \quad (10)$$

According to (10), the precision parameter p_i is increasing in N_i . Therefore, when firm i receives more coverage, the quantity of signals N_i rises, thereby enhancing the precision of the aggregate information signal about the firm. This explains why lower news and analyst coverage is consistent with less accurate information about the firm, which is what we observe for value firms.

As illustrated in the bottom panels of Figure 5, forecasts for value firms also show greater dispersion, indicating higher uncertainty or disagreement among analysts. In addition, value firms are more likely to experience larger earnings surprises, suggesting a gap between expectations and actual earnings. To measure disagreement and earnings surprises, we adopt measures commonly-used in the accounting and finance literature that studies earnings announcements (Livnat and Mendenhall, 2006; Hartzmark and Shue, 2018; Martineau, 2022). Analyst disagreement is quantified by the dispersion of forecasts, calculated as the standard deviation of earnings predictions issued by analysts for a firm's announcement within the 90 days prior to the announcement date, normalized by the absolute value of the consensus forecast. Analyst surprise measures the accuracy of forecasts, defined as the absolute difference between the firm's reported earnings per share for the quarter and the consensus analyst forecast, scaled by the firm's stock price.¹⁸

Higher disagreement for value firms is consistent with lower signal precision. To illustrate this connection, consider two analysts A and B with priors f_A and f_B about the unobserved fundamental f . The two analysts disagree with each other: $f_A - f_B > 0$. If neither of

¹⁷The aggregate signal is derived by averaging the individual signals and then rescaling to ensure that ds_i is a Brownian motion increment.

¹⁸We find similar results using the unscaled earnings surprises.

them receives any new information (signal precision is nil), they are not able to update their beliefs, and the disagreement will therefore still be $f_A - f_B > 0$. On the other extreme, if they both receive a perfectly informative signal about the fundamental (signal precision is perfect), their updated beliefs become $f_A = f$ and $f_B = f$. As a result, the updated disagreement becomes $f_A - f_B = 0$. This illustrates how less precise information is in line with higher disagreement among analysts, which is what we observe for value firms.

Similarly, the higher forecast error for value firms is consistent with lower signal precision. Indeed, if analyst A with prior f_A is unable to update her belief because she observes an uninformative signal, her forecast error is $|f - f_A| > 0$. On the other extreme, if the signal she observes is perfectly informative, she is able to learn the true value of the fundamental, and the forecast error is consequently zero. This scenario underscores how lower quality of information is in line with higher earnings surprises, which is what we observe for value firms.

Building upon our analysis of heterogeneity in information quality, we now turn our attention to firm-level regressions results presented in Table 3. Within our theoretical model, a positive shock to output leads to a larger increase in cashflow expectations for value firms compared to growth firms. In Section 4.1, we confirmed that the correlation between the shock in a firm's expected growth component and the output shock is higher for value firms than for growth firms, using aggregate data from value and growth portfolios. Now, Table 3 further substantiates these findings with firm-level data, demonstrating how individual firms within each category respond to output shocks.

In column (1) of Table 3, we present our main regression specification focusing on value and growth firms only:

$$\begin{aligned} \text{Forecasted growth}_{i,t+1} = & \beta_1 \text{Output Shock}_t + \beta_2 D_{\text{Value},i,t} \\ & + \beta_3 (\text{Output Shock}_t \times D_{\text{Value},i,t}) + \alpha_i + \epsilon_{i,t+1}, \end{aligned} \quad (11)$$

where the dependent variable represents the proportional change in real earnings forecasts. Specifically, it is the relative difference between the earnings forecasted for the upcoming quarter $t + 1$, and the earnings forecasted for the current quarter t . Output Shock_t is the standardized difference between real GDP growth and its forecast, capturing unexpected output growth surprises observed in quarter t . $D_{\text{Value},i,t}$ is a dummy variable indicating whether firm i falls into the high book-to-market (value) in quarter t .

The independent variable of interest, the interaction term, explores if the response to output shocks differs between value and growth firms. The coefficient for this interaction term in column (1) is positive and statistically significant, confirming our model's prediction

that a positive output shock results in a larger upward revision in expectations for value firms' future cashflows compared to growth firms. A one-standard-deviation positive output shock, equivalent to a 1 percentage point increase in real GDP growth beyond the forecasted rate, is associated with a 2.8 percentage point increase in forecasted earnings growth for value firms, as captured by the sum of the coefficients β_1 and β_3 .

In column (2), we investigate the differential effect further by dividing value firms into two subcategories based on the number of signals, which we proxy through analyst coverage. Specifically, value firms are classified as having low analyst coverage if they are at or below the median analyst coverage of value firms in a given year, and high if they are above it. This split is applied only to value firms, so the reference category remains growth firms, consistent with the analysis in column (1). The positive and statistically significant interaction coefficient of 0.084 reported in column (2) indicates that the effect of output shocks on expectations about value firms predominantly originates from those with low analyst coverage.

In columns (3) and (4), we replicate the analyses conducted in columns (1) and (2) but restrict our focus to the half of the sample for which we have RavenPack news data (2000–2022). In this subsample, we use news coverage as the proxy for the number of signals. As before, growth firms remain the reference category in columns (3) and (4). As in column (2), value firms are categorized by news coverage relative to the median news coverage of value firms in a given year in column (4). The results reported in columns (3) and (4) align with those from the earlier analyses in columns (1) and (2): analysts adjust their forecasts upward significantly more for value firms than for growth firms in response to a positive output shock, and this response originates from value firms with lower news coverage.

Finally, in column (5) we estimate the regression model specified in (11) but substitute forecasted earnings growth with realized earnings growth as the dependent variable. Unlike for the forecasts, actual earnings of value and growth firms respond similarly to output shocks. This suggests that the previous finding on the different responses of value versus growth firms forecasts to output shocks is likely explained by expectations shaped by the quality of information signals rather than by actual earnings outcomes.

The differences across book-to-market categories discussed in this section echo the model's implication about the information quality's impact on a firm being either value or growth. In our framework, the valuation of firms is influenced by the precision of available information. Specifically, firms with highly precise information endogenously emerge as growth firms, while those with less precise information are identified as value firms. This conceptual link between information quality and firm characteristics is supported by the empirical evidence in Figure 5 and Table 3. Lower news and analyst coverage, greater forecast disagreement, larger earnings surprises, and significant revisions in analysts' expectations in response to

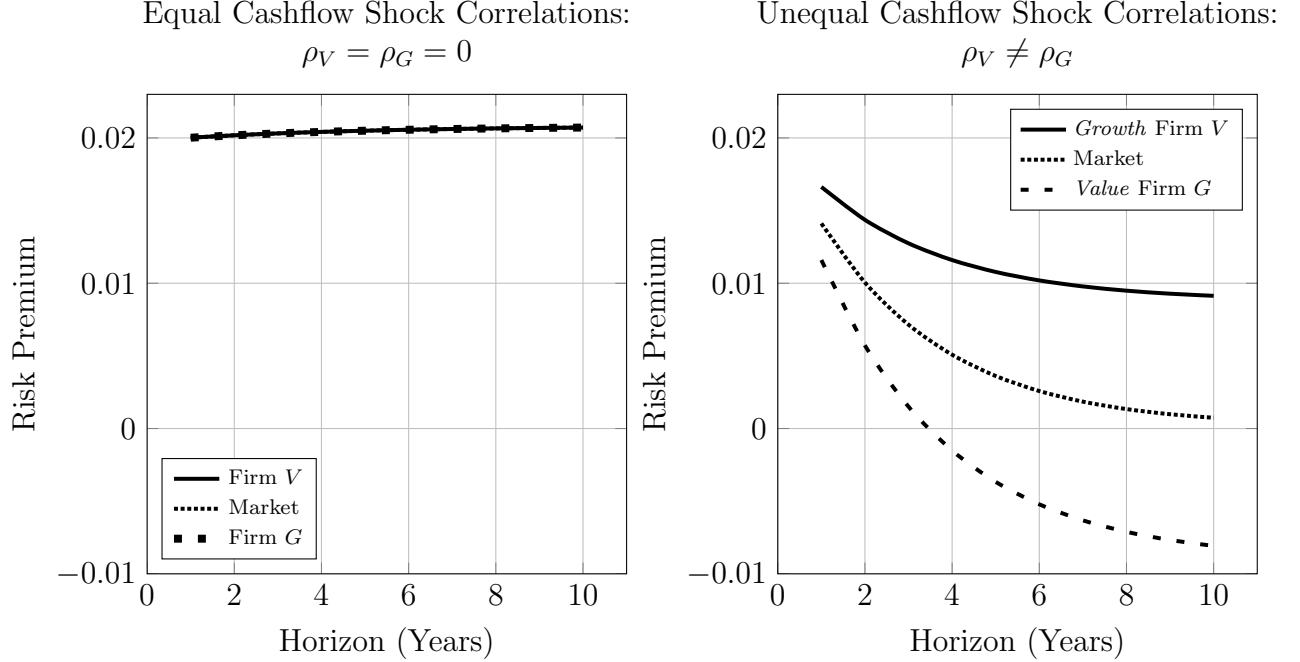
Table 3: Impact of Output Shocks on Earnings Expectations.

We report panel regression results examining the impact of output shock, defined as the standardized difference between real GDP growth and its forecast. In columns (1) to (4), the dependent variable is the difference between the earnings forecasted for the upcoming quarter $t + 1$ and those forecasted for the current quarter t , expressed as a relative change. Column (5) instead examines the actual change in realized earnings over the same period. The dummy variable D_{Value} classifies firms as either high book-to-market (value, $D_{\text{Value}} = 1$) or low book-to-market (growth, $D_{\text{Value}} = 0$). $D_{\text{Value, Low \#Signals}}$ and $D_{\text{Value, High \#Signals}}$ differentiate value firms based on low and high levels of analyst or news coverage in columns (2) and (4), respectively. Coverage is considered low if it is at or below the median coverage of value firms in a given year, and high otherwise. All models include firm fixed effects and the reported intercept is the average value of the fixed effects for the growth firms. Standard errors are clustered at the firm-quarter level. t -statistics are reported below each estimate, with *, **, and *** indicating statistical significance at the 10%, 5%, and 1% levels, respectively.

	Forecasted Growth				Realized Growth
	(1)	(2)	(3)	(4)	(5)
Output Shock	-0.019 (-0.79)	-0.019 (-0.79)	-0.038 (-1.00)	-0.038 (-1.00)	0.019 (1.50)
D_{Value}	-0.046 (-1.40)		0.050 (1.02)		-0.070* (-1.73)
Output Shock $\times$ D_{Value}	0.047*** (2.77)		0.060** (2.16)		-0.016 (-0.95)
$D_{\text{Value, Low \#Signals}}$		-0.080** (-2.06)		0.052 (0.81)	
$D_{\text{Value, High \#Signals}}$		-0.011 (-0.26)		0.048 (0.74)	
Output Shock $\times$ $D_{\text{Value, Low \#Signals}}$		0.084*** (3.99)		0.086** (2.52)	
Output Shock $\times$ $D_{\text{Value, High \#Signals}}$		0.013 (0.65)		0.036 (1.17)	
Constant	0.037*** (3.15)	0.037*** (3.13)	-0.001 (-0.06)	-0.001 (-0.06)	-0.005 (-0.33)
Firm FE	Yes	Yes	Yes	Yes	Yes
N obs.	173,841	173,841	95,483	95,483	174,346
R^2	0.050	0.050	0.055	0.055	0.073

Figure 6: Equity Term Structures in the Full-Information Economy.

This figure depicts the model-implied term structure of risk premium for firm V (solid line), firm G (dashed line), and the market (dotted line). In the left panel, cashflow shock correlations are equal: $\rho_V = \rho_G = 0$. In the right panel, cashflow shock correlations are different across the two firms $\rho_V \neq \rho_G$. To insure finite prices, the subjective discount factor is $\delta = 0.93$. The other parameters are provided in Section 4.1.



output shocks suggest that investors face a relative lack of precise information about the future cashflows of value firms. In contrast, growth firms, characterized by higher news and analyst coverage, less forecast disagreement, and smaller earnings surprises, appear to have higher quality information available to investors.

4.4 Discussion on the Full-Information Economy

This section discusses the results obtained in the full-information economy, defined as the economy in which the agent observes the expected output growth component x , firm V expected growth component x_V , and firm G expected growth component x_G . In the full-information economy, the agent does not need to learn about any variable.

Let us first assume that the cashflow shock correlations of firms V and G are equal: $\rho_V = \rho_G = 0$. In contrast to what is obtained in the partial-information economy with equal cashflow shock correlations (see Section 4.2.1), the dividend yields of firms V and G are both equal to 0.23% in this full-information economy. Therefore, firm V cannot be

interpreted as a value firm, and firm G cannot be interpreted as growth firm. In addition, the risk premium of firm V is equal to that of firm G , and consequently equal to that of the market as well. The market risk premium is equal 2.08%, which is considerably lower than the market risk premium of 7.37% obtained in the partial-information economy with $\rho_V = \rho_G = 0$. In the full-information economy, the value premium is equal to zero, whereas it is 3.26% in the partial-information economy with $\rho_V = \rho_G = 0$. Unsurprisingly, the equity term structures of firm V , firm G , and the market are equal in the full-information economy with $\rho_V = \rho_G = 0$ (see Figure 6, left panel). These results hold because, in the full-information economy with equal cashflow shock correlations, firm V expected growth component x_V is perfectly symmetric to firm G expected growth component x_G ; they feature the same parameters and unpriced, independent shocks.

Let us now consider the full-information economy, in which firms V and G have different cashflow shock correlations: $\rho_V \neq \rho_G$. In this economy, the model-implied dividend yield of firm V is 0.79%, whereas that of firm G is 1.69%. That is, firm V is a *growth firm*, and firm G is a *value firm* in the full-information economy. The risk premium of the growth firm V is 0.90%, and the risk premium of the value firm G is -0.78% . Therefore, the value premium is negative and equal to -1.68% in the full-information economy. The model-implied market risk premium is 0.36%, substantially lower than the market risk premium of 6.82% obtained in the corresponding partial-information economy. The right panel of Figure 6 shows that the equity term structure of the *value firm* G is markedly downward-sloping, that of the *growth firm* V is slightly downward-sloping, and that of the market is moderately downward-sloping. All of these results are inconsistent with what is observed in the data (Fama and French, 1993; Giglio et al., 2024).

The economic mechanism explaining these results is as follows. The substantial, negative cashflow shock correlation of firm G implies that firm G long-term cashflows are less risky than short-term cashflows. As a result, the risk premium required to hold firm G long-term dividend strips is substantially lower than that required to hold firm G short-term dividend strips. Similarly, firm V long-term dividend strip risk premiums are slightly lower than firm V short-term dividend strip risk premiums because the cashflow shock correlation of firm V is slightly negative. Furthermore, the agent requires a higher risk premium to hold assets having cashflow risk concentrated in the longer term because the agent has a preference for early resolution of uncertainty. Since firm G cashflow risk is less concentrated in the long term than firm V cashflow risk, the risk premium of firm G is lower than that of firm V .

Finally, firm G 's dividend yield is higher than firm V 's dividend yield for the following reason. The substantially negative cashflow shock correlation of firm G caps the expected future cashflow growth of firm G relative to firm V particularly strongly. In the full-information

economy, the lower expected cashflow growth effect of firm G is strong enough to dominate its lower discounting effect implied by its lower risk premium. This explains the lower valuations of firm G relative to firm V ; firm G 's dividend yield is higher than firm V 's dividend yield.

4.5 Discussion on CAPM-Betas and the Value Premium

Whether or not CAPM-betas can explain the value premium has been debated in the empirical literature. For instance, [Fama and French \(1993, 2006\)](#) argue that the CAPM does not explain the value premium, whereas [Ang and Chen \(2007\)](#) show that it explains it.

In the model with partial information and different signal precisions only (Section 4.2.1), the CAPM-beta of the value firm V is 0.59, and that of the growth firm G is 1.04. That is, the CAPM-beta of the value firm is lower than that of the growth firm, whereas the risk premium of the value firm is about 3.3% larger than that of the growth firm. This shows that the CAPM does not explain the value premium in this model.

To understand this result, consider the structure of the market's cashflows: Both the value and growth firms cashflows contribute to the market cashflow, which is consequently influenced by two distinct unpriced sources of risk, s_V and s_G . Given that the signal about firm V has a lower precision than the signal about firm G , the exposure of firm V 's cashflow to unpriced signal risks is lower than that of firm G 's cashflow. This reduced exposure contributes to the observed CAPM-beta discrepancy between the two firms. Yet, the risk premium of the value firm V exceeds that of the growth firm G . As discussed in Section 4.2.1, this difference is attributed to investors perceiving the expected growth shocks of firm V to be more strongly correlated with priced shocks.

While the learning model with different signal precisions accounts for about 80% of the observed value premium, incorporating heterogeneity in cashflow shock correlations (Section 4.2.2) yields a model-implied value premium of 3.9%, corresponding to 95% of the observed value premium. Again here, the value premium is not explained by the CAPM. Indeed, the CAPM-beta of the value firm V is 1.04, and that of the growth firm G is 0.97. These CAPM-betas imply a CAPM-implied value premium of 0.48% only, far below the actual model-implied value premium of 3.9%.

4.6 Impact of Time-Varying Signal Precision and Cashflow Shock Correlation on the Value Premium

In Section 2.2, the parameters driving the cashflow dynamics of firms V and G were estimated using earnings data on a portfolio of value firms and a portfolio of growth firms, respectively.

Firms V and G were consequently interpreted as representative value and growth firms, i.e., firms representing a large basket of value and growth firms.

In contrast, firms V and G are interpreted as individual firms in the current section. The reason is that firms V and G now feature time-varying signal precision and cashflow shock correlation, which drive price-dividend ratios in equilibrium. As a result, individual firms might not necessarily belong permanently to the value or growth categories. Extending the model provided in Section 2.2, firm V -specific and firm G -specific expected growth components are assumed to have the following dynamics

$$\begin{aligned} d\hat{x}_{V,t} &= -\bar{\lambda}\hat{x}_{V,t}dt + \hat{\sigma}_{x_{V,t}}d\hat{B}_t + p_{V,t}\bar{\sigma}\sqrt{1 - \rho_{V,t}^2}ds_{V,t}, \text{ with} \\ p_{V,t} &= \pi_t p_{min} + (1 - \pi_t)p_{max} \text{ and } \rho_{V,t} = \pi_t \rho_{max} + (1 - \pi_t)\rho_{min}, \end{aligned} \quad (12)$$

$$\begin{aligned} d\hat{x}_{G,t} &= -\bar{\lambda}\hat{x}_{G,t}dt + \hat{\sigma}_{x_{G,t}}d\hat{B}_t + p_{G,t}\bar{\sigma}\sqrt{1 - \rho_{G,t}^2}ds_{G,t}, \text{ with} \\ p_{G,t} &= \pi_t p_{max} + (1 - \pi_t)p_{min} \text{ and } \rho_{G,t} = \pi_t \rho_{min} + (1 - \pi_t)\rho_{max}, \end{aligned} \quad (13)$$

$$d\pi_t = \lambda_\pi(\bar{\pi} - \pi_t)dt + \sigma_\pi\pi_t(1 - \pi_t)dW_t, \quad (14)$$

where $p_{min} = p_V - e$, $p_{max} = p_G + e$, $\rho_{min} = \rho_G - e$, $\rho_{max} = \rho_V + e$, $e \geq 0$ is a small constant defining the neighborhood of the parameter values $(p_V, p_G, \rho_V, \rho_G)$, and for $i \in \{V, G\}$

$$\begin{aligned} \hat{\sigma}_{x_{i,t}} &\equiv \frac{\rho_{i,t}\bar{\sigma}(\sigma_D - \sigma_C) + \gamma_{x_{i,t}}}{\sigma_D - \sigma_C}, \\ \gamma_{x_{i,t}} &= (\sigma_D - \sigma_C) \left(-\rho_{i,t}\bar{\sigma} - \bar{\lambda}(\sigma_D - \sigma_C) \right. \\ &\quad \left. + \sqrt{(1 + p_{i,t}^2(-1 + \rho_{i,t}^2))\bar{\sigma}^2 + \bar{\lambda}^2(\sigma_D - \sigma_C)^2 + 2\bar{\lambda}\rho_{i,t}\bar{\sigma}(\sigma_D - \sigma_C)} \right). \end{aligned}$$

The Brownian motion W_t driving π_t is independent of the Brownian motions $\hat{B}_t$, $s_{V,t}$, and $s_{G,t}$. The dynamics of the other variables, $(d \log C, d \log D_V, d \log D_G, d\hat{x})$, remain the same as in Section 2.2.

A parallel can be drawn between the dynamics in (14) and the dynamics obtained when filtering an unobservable continuous-time 2-state Markov chain $\tilde{s}_t$ taking values in, say, $\{0, 1\}$. Indeed, the dynamics of π_t in (14) are equivalent to the dynamics of the conditional (filtered) probability that $\tilde{s}_t = 1$ in Veronesi (1999). The long-term mean $\bar{\pi} \in (0, 1)$ in (14) is interpreted as the long-term (unconditional) probability that $\tilde{s}_t = 1$, which is also a measure of persistence of state $\{1\}$. The parameter $\lambda_\pi > 0$ measures the speed at which π_t

reverts towards $\bar{\pi}$, and the diffusion term $\pi_t(1 - \pi_t)$ insures that π_t remains between 0 and 1. Equations (12) and (13) show that the signal precisions $(p_{V,t}, p_{G,t})$ and cashflow shock correlations $(\rho_{V,t}, \rho_{G,t})$ are time-varying π_t -weighted-averages of the parameters (p_{min}, p_{max}) and (ρ_{min}, ρ_{max}) , respectively. When π_t is close to 1, $p_{V,t}$ and $\rho_{V,t}$ are in the neighborhood of p_V and ρ_V , respectively, whereas $p_{G,t}$ and $\rho_{G,t}$ are in the neighborhood of p_G and ρ_G , respectively. Conversely, when π_t is close to 0, $p_{V,t}$ and $\rho_{V,t}$ are in the neighborhood of p_G and ρ_G , respectively, whereas $p_{G,t}$ and $\rho_{G,t}$ are in the neighborhood of p_V and ρ_V , respectively. Setting the long-term probability $0.5 < \bar{\pi} < 1$ insures that firm V 's signal precision $p_{V,t}$ and cashflow shock correlation $\rho_{V,t}$ persist more in the neighborhood of the parameter values p_V and ρ_V , respectively, than in the neighborhood of p_G and ρ_G . Similarly, $0.5 < \bar{\pi} < 1$ also insures that firm G 's signal precision $p_{G,t}$ and cashflow shock correlation $\rho_{G,t}$ persist more in the neighborhood of the parameter values p_G and ρ_G , respectively, than in the neighborhood of p_V and ρ_V .

To summarize, firm V 's (resp., firm G 's) signal precision and cashflow shock correlation persist in the neighborhood of the parameter values p_V and ρ_V (resp., p_G and ρ_G), respectively, but sometimes depart and come near to the parameter values p_G and ρ_G (resp., p_V and ρ_V), respectively. This model reflects the fact that individual firms might not necessarily belong permanently to the value or growth categories. Indeed, belonging to a given category might be partially transitory.

The mean-reversion speed and volatility parameters of π_t are set to $\lambda_\pi = 0.5$ and $\sigma_\pi = 2$, respectively. The long-term probability $\bar{\pi}$ and the neighborhood-defining constant e are set to different values belonging to the intervals $(0.5, 1)$ and $[0, 0.1]$, respectively. The other parameter values are provided in Section 4.1. As in Section 2.3, firm V and G stock prices are obtained by log-linearizing the corresponding price-dividend ratios.

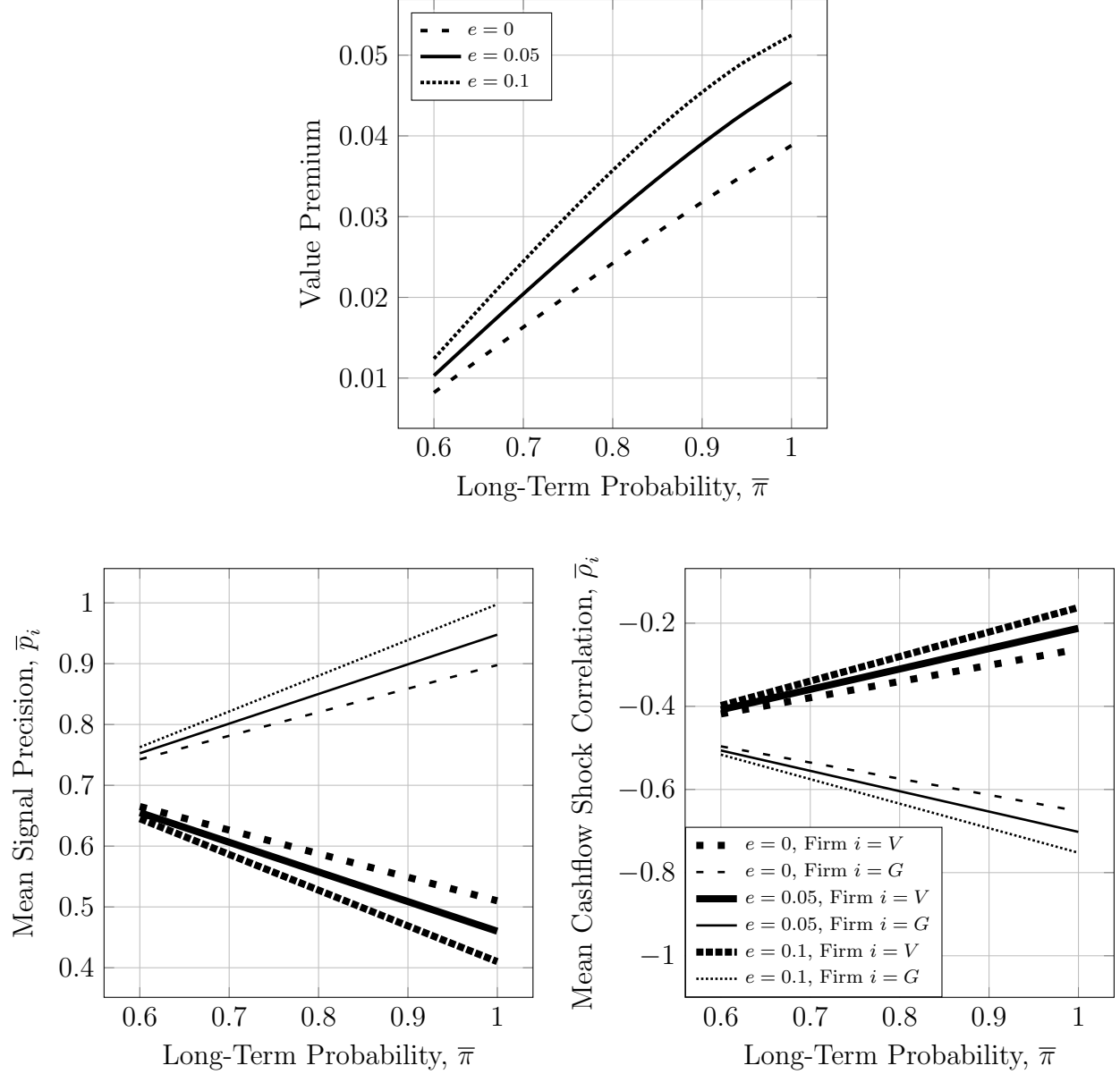
Figure 7 depicts the value premium (top panel) obtained for different values of the long-term probability $\bar{\pi}$ and neighborhood-defining constant e . The bottom left panel depicts the corresponding long-term signal precision means of firms V and G : $\bar{p}_V \equiv \lim_{t \rightarrow \infty} \mathbb{E}(p_{V,t}) = \bar{\pi}p_{min} + (1 - \bar{\pi})p_{max}$ and $\bar{p}_G \equiv \lim_{t \rightarrow \infty} \mathbb{E}(p_{G,t}) = \bar{\pi}p_{max} + (1 - \bar{\pi})p_{min}$. The bottom right panel plots the long-term cashflow shock correlation means of firms V and G : $\bar{\rho}_V \equiv \lim_{t \rightarrow \infty} \mathbb{E}(\rho_{V,t}) = \bar{\pi}\rho_{max} + (1 - \bar{\pi})\rho_{min}$ and $\bar{\rho}_G \equiv \lim_{t \rightarrow \infty} \mathbb{E}(\rho_{G,t}) = \bar{\pi}\rho_{min} + (1 - \bar{\pi})\rho_{max}$.

In the limiting case where the long-term probability $\bar{\pi}$ is equal to 1, the process π_t is equal to 1 for all t ,¹⁹ implying that $(p_{V,t}, \rho_{V,t}, p_{G,t}, \rho_{G,t}) = (p_{min}, \rho_{max}, p_{max}, \rho_{min}) \stackrel{\text{if } e=0}{=} (p_V, \rho_V, p_G, \rho_G)$ for all t . That is, firms V and G belong permanently to respectively the value and growth categories, as in Section 4.2.2. Therefore, the value premium obtained in the limiting case with $e = 0$ corresponds to the 3.9% value premium obtained in the constant-

¹⁹Note that π_t is assumed to start at its long-term mean: $\pi_0 = \bar{\pi}$.

Figure 7: Value Premium with Time-Varying Signal Precision and Cashflow Shock Correlation.

This figure depicts the value premium (top), the mean signal precision of firms V and G (bottom left), and the mean cashflow shock correlation of firms V and G (bottom right) as a function of the long-term probability $\bar{\pi}$. In the bottom panels, the thick curves are for firm V , and the thin curves are for firm G . The process π is set to its long-term mean $\bar{\pi}$. The model parameters are provided in Section 4.1.



precision, constant-correlation model of Section 4.2.2. When the long-term probability $\bar{\pi}$ is smaller than 1, firms V and G do not permanently belong to the value and growth categories anymore. Indeed, as $\bar{\pi}$ decreases, firm V 's (resp., firm G 's) likelihood to remain a value firm (resp., growth firm) at each point in time decreases. In other words, as $\bar{\pi}$ decreases, firm V (resp., firm G) is less frequently classified as a value firm (resp., growth firm) and more frequently classified as a growth firm (resp., value firm); firms V and G become more and

more alike. As a result, the mean signal precisions and mean cashflow shock correlations of firms V and G get closer to each other (Figure 7, bottom panels), and the value premium decreases (Figure 7, top panel). The value premium computed among two individual firms V and G belonging non-permanently to the value and growth categories is about 1% when $\bar{\pi} = 0.6$, 2% when $\bar{\pi} = 0.7$, 3% when $\bar{\pi} = 0.8$, and 4% when $\bar{\pi} = 0.9$.

5 Conclusion

This paper shows that learning about expected future cashflows generates a value premium, an upward-sloping equity term structure for value firms, and a flat equity term structure for growth firms. More accurate information on a firm’s future cashflow means less risk, and therefore higher valuations and a lower risk premium in equilibrium. That is, firms featuring accurate information endogenously emerge as growth firms, whereas firms featuring less accurate information endogenously emerge as value firms. Moreover, learning using accurate information implies that risk is low in the long term, whereas learning using less accurate information yields high risk in the long term. This explains why the model-implied term structure of risk premium is flat for growth firms and upward sloping for value firms. Using earnings, returns, and news data on value and growth firms, we show that the predictions of the model are consistent with what is observed in the data.

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Appendix

A Proofs

A.1 Proposition 1

Proof. The filtering equations follow from standard results in non-linear filtering theory, see [Liptser and Shiryaev \(2001\)](#). Since the posterior variance-covariance matrix is a deterministic function of time, we replace it with its steady-state value, as in [Scheinkman and Xiong \(2003\)](#) and [Dumas et al. \(2009\)](#) among others. That is, we assume that the agent has already observed a long enough history of output growth rates to reach the stationary variance-covariance matrix. The corresponding steady-state volatility of the filter is denoted by $\hat{\sigma}_x$.

The steady-state volatility $\hat{\sigma}_x$ satisfies

$$\hat{\sigma}_x = \frac{\gamma_x}{\sigma_C},$$

where the steady-state posterior variance $\gamma_x \equiv \lim_{t \rightarrow \infty} \text{Var}[x_t | \mathcal{F}_t^o]$ solves the following equation

$$0 = \sigma_x^2 - 2\lambda_x \gamma_x - \frac{\gamma_x^2}{\sigma_C^2}.$$

The solution is

$$\gamma_x = \sigma_C \left(-\lambda_x \sigma_C + \sqrt{\lambda_x^2 \sigma_C^2 + \sigma_x^2} \right).$$

□

A.2 Proposition 2

Proof. To project firm i 's cashflow D_i on the observable filtration, we define the process H_i as

$$H_{i,t} \equiv \log D_{i,t} - \log C_t.$$

The drift of H_i depends exclusively on the unobservable component x_i . Indeed,

$$dH_{i,t} = x_{i,t} dt + (\sigma_D - \sigma_C) dB_t.$$

Following [Liptser and Shiryaev \(2001\)](#), observing H_i and s_i yields the agent's estimate of x_i , $\hat{x}_{i,t} \equiv \mathbb{E}[x_{i,t} | \mathcal{F}_t^o]$. The dynamics of the filter $\hat{x}_i$ are given by

$$d\hat{x}_{i,t} = -\bar{\lambda} \hat{x}_{i,t} dt + \hat{\sigma}_{x_i} d\hat{B}_t + p_i \bar{\sigma} \sqrt{1 - \rho_i^2} ds_{i,t}$$

where $\hat{\sigma}_{x_i} \equiv \frac{\rho_i \bar{\sigma}(\sigma_D - \sigma_C) + \gamma_{x_i}}{\sigma_D - \sigma_C}$ and $\gamma_{x_i} \equiv \lim_{t \rightarrow \infty} \text{Var}[x_{i,t} | \mathcal{F}_t^o]$ is the steady-state posterior variance. The dynamics of H_i with respect to the observable filtration is

$$dH_{i,t} = \hat{x}_{i,t} dt + (\sigma_D - \sigma_C) d\hat{B}_t,$$

and the $\mathcal{F}^o$ -adapted dynamics of D_i follow directly from the definition of H_i . Note that the Brownian motion $\hat{B}$ driving the dynamics of D_i and $\hat{x}_i$ is exactly the same as that driving the output C and the output expected growth component $\hat{x}$ in Equations (6) and (7), respectively.

The steady-state posterior variance γ_{x_i} satisfies

$$0 = (1 - p_i^2)(1 - \rho_i^2)\bar{\sigma}^2 - \frac{(2(\sigma_D - \sigma_C)(\rho_i \bar{\sigma} + \bar{\lambda}(\sigma_D - \sigma_C)) + \gamma_{x_i})\gamma_{x_i}}{(\sigma_D - \sigma_C)^2}.$$

The solution is

$$\begin{aligned} \gamma_{x_i} = & (\sigma_D - \sigma_C) \left(-\rho_i \bar{\sigma} - \bar{\lambda}(\sigma_D - \sigma_C) \right. \\ & \left. + \sqrt{(1 + p_i^2(-1 + \rho_i^2))\bar{\sigma}^2 + \bar{\lambda}^2(\sigma_D - \sigma_C)^2 + 2\bar{\lambda}\rho_i \bar{\sigma}(\sigma_D - \sigma_C)} \right). \end{aligned}$$

□

A.3 Corollary 1

Proof. Using the expression for γ_{x_i} we have

$$\begin{aligned} \hat{\sigma}_{x_i} & \equiv \rho_i \bar{\sigma} + \frac{\gamma_{x_i}}{\sigma_D - \sigma_C} \\ & = -\bar{\lambda}(\sigma_D - \sigma_C) + \sqrt{(1 + p_i^2(-1 + \rho_i^2))\bar{\sigma}^2 + \bar{\lambda}^2(\sigma_D - \sigma_C)^2 + 2\bar{\lambda}\rho_i \bar{\sigma}(\sigma_D - \sigma_C)}. \end{aligned}$$

Differentiating $\hat{\sigma}_{x_i}$ with respect to p_i :

$$\frac{\partial \hat{\sigma}_{x_i}}{\partial p_i} = -\frac{p_i(1 - \rho_i^2)\bar{\sigma}^2}{\sqrt{(1 + p_i^2(-1 + \rho_i^2))\bar{\sigma}^2 + \bar{\lambda}^2(\sigma_D - \sigma_C)^2 + 2\bar{\lambda}\rho_i \bar{\sigma}(\sigma_D - \sigma_C)}} \leq 0$$

with strict inequality if $p_i \neq 0$ and $\rho_i \neq \pm 1$.

□

A.4 Corollary 2

Proof. The conditional variance of the filter in Equation (8) is given by:

$$\frac{1}{dt} \text{Var}(d\hat{x}_{i,t}) = \hat{\sigma}_{x_i}^2 + p_i^2 \bar{\sigma}^2 (1 - \rho_i^2).$$

Taking the partial derivative with respect to p_i , we obtain:

$$2\widehat{\sigma}_{x_i} \frac{\partial \widehat{\sigma}_{x_i}}{\partial p_i} + 2p_i \bar{\sigma}^2 (1 - \rho_i^2) = \frac{2p_i \bar{\lambda} (1 - \rho_i^2) \bar{\sigma}^2 (\sigma_D - \sigma_C)}{\sqrt{(1 + p_i^2 (-1 + \rho_i^2)) \bar{\sigma}^2 + \bar{\lambda}^2 (\sigma_D - \sigma_C)^2 + 2\bar{\lambda} \rho_i \bar{\sigma} (\sigma_D - \sigma_C)}}.$$

□

A.5 Proposition 3

Proof. This proof follows closely [Eraker and Shaliastovich \(2008\)](#). We conjecture that the log wealth-consumption ratio is affine in the state variables $X_t = (\widehat{y}_t, \widehat{y}_{V,t}, \widehat{y}_{G,t}, \widehat{x}_t, \widehat{x}_{V,t}, \widehat{x}_{G,t})^\top$, where $\widehat{y}_t \equiv \log C_t$ and $\widehat{y}_{i,t} \equiv \log D_{i,t}$, so that

$$wc_t \equiv \log \frac{W_t}{C_t} = A_0 + A_1^\top X_t, \quad (15)$$

and use the fact that the state variables belong to the affine class, so that their dynamics can be written as:

$$\begin{aligned} dX_t &= \mu(X_t)dt + \Sigma(X_t)d\widehat{B}_t \\ \mu(X_t) &= \mathcal{M} + \mathcal{K}X_t \\ \Sigma(X_t)\Sigma(X_t)^\top &= h + \sum_{i=1}^6 H^i X_t^i, \end{aligned}$$

where X_t is the vector of the filtered state variables, $\Sigma(X_t) \in \mathbb{R}^{6 \times 1}$ encodes the diffusions of the state variables, $\mathcal{M} \in \mathbb{R}^6$, $\mathcal{K} \in \mathbb{R}^{6 \times 6}$, $h \in \mathbb{R}^{6 \times 6}$, $H \in \mathbb{R}^{6 \times 6 \times 6}$, and $\widehat{B}_t$ is a standard Brownian motion.

The dynamics of the state-price density then can be written as

$$d \log M_t = (\theta \log \delta - (\theta - 1) \log k_1 + (\theta - 1)(k_1 - 1)A_1'(X_t - \mu_X))dt - \Omega' dX_t, \quad (16)$$

where $\mu_X = (0, 0, 0, 0, 0, 0)^\top$, $\Omega = \gamma(1, 0, 0, 0, 0, 0)^\top + (1 - \theta)k_1 A_1$, and the coefficients $A_0 \in \mathbb{R}$ and $A_1 \in \mathbb{R}^6$ are the loadings defined in (15).

The coefficients $A_0 \in \mathbb{R}$, $A_1 \in \mathbb{R}^6$ solve the following system of equations

$$0 = \mathcal{K}^\top \chi - \theta(1 - k_1)A_1 + \frac{1}{2}\chi^\top H \chi, \quad (17)$$

$$0 = \theta(\log \delta + k_0 - (1 - k_1)A_0) + \mathcal{M}^\top \chi + \frac{1}{2}\chi^\top h \chi, \quad (18)$$

and the linearization coefficient $k_1 \in \mathbb{R}$ satisfies

$$\theta \log k_1 = \theta(\log \delta + (1 - k_1)A_1^\top \mu_X) + \mathcal{M}^\top \chi + \frac{1}{2}\chi^\top h \chi,$$

where $\chi = \theta \left((1 - \frac{1}{\psi})(1, 0, 0, 0, 0, 0)^\top + k_1 A_1 \right)$.

Solving (17) for the vector of loadings $A_1 \in \mathbb{R}^6$ gives

$$A_1^\top = \left(0, 0, 0, \frac{1 - 1/\psi}{1 - k_1(1 - \lambda_x)}, 0, 0 \right).$$

Plugging this solution in equation (18) allows to solve for the coefficient A_0 .

From the arbitrage theory we know that the state-price density M_t satisfies

$$\frac{dM_t}{M_t} = -r_t dt - \Lambda_t^\top dB_t.$$

where r_t is the risk-free rate and Λ_t is the market price of risk vector.

Eraker and Shaliastovich (2008) show that from the expression for the state price density in (16), the risk free rate and market price of risk vector can be determined as follows:

$$\begin{aligned} r_t &= r_0 + r_1^\top X_t, \\ \Lambda_t &= \Sigma(X_t)^\top \Omega, \end{aligned}$$

where the vector $\Omega = \gamma(1, 0, 0, 0, 0, 0)^\top + (1 - \theta)k_1 A_1$ and the coefficients $r_0 \in \mathbb{R}$ and $r_1 \in \mathbb{R}^6$ solve the system of equations

$$\begin{aligned} r_1 &= (1 - \theta)(k_1 - 1)A_1 + \mathcal{K}^\top \Omega - \frac{1}{2} \Omega^\top H \Omega, \\ r_0 &= -\theta \log \delta + (\theta - 1)(\log k_1 + (k_1 - 1)A_1^\top \mu_X) + \mathcal{M}^\top \Omega - \frac{1}{2} \Omega^\top h \Omega. \end{aligned}$$

Solving for r_1, r_0 gives $r_1^\top = (0, 0, 0, 1/\psi, 0, 0)$ and

$$r_0 = -\frac{1 - \gamma}{1 - 1/\psi} \log \delta + \frac{1/\psi - \gamma}{1 - 1/\psi} \log k_1 + \gamma \mu - \frac{1}{2} \Theta(\hat{\sigma}_x),$$

where

$$\Theta(\hat{\sigma}_x) \equiv \frac{(k_1 \hat{\sigma}_x - \gamma(\sigma_C + k_1(\hat{\sigma}_x + (\lambda_x - 1)\sigma_C))\psi)^2}{\psi^2(1 + k_1(\lambda_x - 1))^2},$$

with $\hat{\sigma}_x$ is defined in Proposition 1. Similarly, market price of risk can be written as

$$\Lambda = \gamma \hat{\sigma}_y + \left(\frac{(\gamma - 1/\psi)}{1/k_1 - (1 - \lambda_x)} \right) \hat{\sigma}_x.$$

Finally, following Eraker and Shaliastovich (2008), the dynamics of the vector of state variables X_t under the risk neutral measure $\mathbb{Q}$ are given by

$$dX_t = (\mathcal{M}^\mathbb{Q} + \mathcal{K}^\mathbb{Q} X_t) dt + \Sigma(X_t) d\hat{B}_t^\mathbb{Q},$$

where $\hat{B}_t^\mathbb{Q} = \hat{B}_t + \int_0^t \Lambda_s ds$ is a $\mathbb{Q}$ -Brownian motion and the coefficients $\mathcal{M}^\mathbb{Q} \in \mathbb{R}^6$ and

$\mathcal{K}^{\mathbb{Q}} \in \mathbb{R}^{6 \times 6}$ satisfy

$$\mathcal{M}^{\mathbb{Q}} = \mathcal{M} - h\Omega, \quad (19)$$

$$\mathcal{K}^{\mathbb{Q}} = \mathcal{K} - H\Omega. \quad (20)$$

□

A.6 Proposition 4

Proof. The price of a dividend strip satisfies

$$S_i(t, \tau) = \mathbb{E}_t^{\mathbb{Q}} \left(e^{-\int_t^{t+\tau} r_s ds} D_{i,t+\tau} \right) = e^{w_{0i}(\tau) + w_{1i}(\tau)X_t}, \text{ with } i \in \{V, G\}$$

and $X_t = (\hat{y}_t, \hat{y}_{V,t}, \hat{y}_{G,t}, \hat{x}_t, \hat{x}_{V,t}, \hat{x}_{G,t})^\top$. Eraker and Shaliastovich (2008) show that the functions $w_{0i}(\tau) \in \mathbb{R}$ and $w_{1i}(\tau) \in \mathbb{R}^6$ solve the following system of Ricatti equations

$$\frac{\partial}{\partial \tau} w_{1i}(\tau) = -r_1 + \mathcal{K}^{\mathbb{Q}\top} w_{1i}(\tau) + \frac{1}{2} w_{1i}(\tau)^\top H w_{1i}(\tau), \quad (21)$$

$$\frac{\partial}{\partial \tau} w_{0i}(\tau) = -r_0 + \mathcal{M}^{\mathbb{Q}\top} w_{1i}(\tau) + \frac{1}{2} w_{1i}(\tau)^\top h w_{1i}(\tau), \quad (22)$$

with boundary conditions $w_{0j}(0) = 0$ and either $w_{1V}(0) = (0, 1, 0, 0, 0, 0)^\top$ or $w_{1G}(0) = (0, 0, 1, 0, 0, 0)^\top$. Coefficients $\mathcal{M}^{\mathbb{Q}} \in \mathbb{R}^6$ and $\mathcal{K}^{\mathbb{Q}} \in \mathbb{R}^{6 \times 6}$ are characterized in (19)–(20).

Solving (21) gives

$$w_{1V}(\tau)^\top = \left(0, 1, 0, \frac{1}{\lambda_x \psi} (1 - e^{-\lambda_x \tau}) (\psi - 1), \frac{1}{\bar{\lambda}} (1 - e^{-\bar{\lambda} \tau}), 0 \right),$$

$$w_{1G}(\tau)^\top = \left(0, 0, 1, \frac{1}{\lambda_x \psi} (1 - e^{-\lambda_x \tau}) (\psi - 1), 0, \frac{1}{\bar{\lambda}} (1 - e^{-\bar{\lambda} \tau}) \right).$$

Using these results in (22) allows to solve for the functions w_{0i} , $i \in \{V, G\}$. The risk premium of asset i dividend strip for with time to maturity τ is computed as

$$\text{RP}_i(t, \tau) = -\frac{1}{dt} \left\langle \frac{dM_t}{M_t}, \frac{dS_i(t, \tau)}{S_i(t, \tau)} \right\rangle.$$

□

A.7 Corollary 3

Proof. We start by noting that as τ approaches 0, the dividend strip risk premium for firm i , $\text{RP}_i(\tau)$, converges to $\sigma_D \Lambda$. Therefore, the difference between $\text{RP}_i(\tau)$ and its limit as τ approaches 0 is given by the expression $(w_{\hat{x}}(\tau) \hat{\sigma}_x + w_{\hat{x}_i}(\tau) \hat{\sigma}_{x_i}) \Lambda$.

Differentiating this difference with respect to the precision of the signal, p_i , yields

$$w_{\hat{x}_i}(\tau)\Lambda\frac{\partial\hat{\sigma}_{x_i}}{\partial p_i}.$$

According to Corollary 1, $\hat{\sigma}_{x_i}$ decreases as p_i increases. Given that $w_{\hat{x}_i}(\tau)$ is positive for any $\tau > 0$, and assuming $\Lambda > 0$, the derivative of the risk premium difference with respect to p_i is negative.

□

A.8 Proposition 5

Proof. Following Eraker and Shaliastovich (2008) we consider an approximate equilibrium solution for the price-dividend ratios, which are obtained, as wealth-consumption ratio in Proposition 3, through the log-linearization of returns. Namely, the log price–dividend ratio of firm i is linear in the state variables,

$$pd_{i,t} \equiv \log \frac{P_{i,t}}{D_{i,t}} = A_{0i} + A_{1i}^\top X_t, \text{ with } i \in \{V, G\}$$

and $X_t = (\hat{y}_t, \hat{y}_{Vt}, \hat{y}_{Gt}, \hat{x}_t, \hat{x}_{Vt}, \hat{x}_{Gt})^\top$. The coefficients $A_{0i} \in \mathbb{R}$, $A_{1i} \in \mathbb{R}^6$ solve the following system of equations

$$0 = \mathcal{K}^\top \chi_i + (\theta - 1)(k_1 - 1)A_1 + (k_{1i} - 1)A_{1i} + \frac{1}{2}\chi_i^\top H \chi_i, \quad (23)$$

$$\begin{aligned} 0 = & \theta \ln \delta - (\theta - 1) (\ln k_1 + (k_1 - 1)A_1^\top \mu_X) - (\ln k_{1i} + (k_{1i} - 1)A_{1i}^\top \mu_X) \\ & + \mathcal{M}^\top \chi_i + \frac{1}{2}\chi_i^\top h \chi_i, \end{aligned} \quad (24)$$

where $\chi_V = (0, 1, 0, 0, 0, 0)^\top + k_{1V}A_{1V} - \Omega$, $\chi_G = (0, 0, 1, 0, 0, 0)^\top + k_{1G}A_{1G} - \Omega$, $k_{1V} \in \mathbb{R}$ and $k_{1G} \in \mathbb{R}$ are the linearization coefficients for the stock return.

Solving (23) for the vector of loadings $A_{1i} \in \mathbb{R}^6$ gives

$$\begin{aligned} A_{1V}^\top &= \left(0, 0, 0, \frac{1 - 1/\psi}{1 - k_{1V}(1 - \lambda_x)}, \frac{1}{1 - k_{1V}(1 - \bar{\lambda})}, 0\right) \\ A_{1G}^\top &= \left(0, 0, 0, \frac{1 - 1/\psi}{1 - k_{1G}(1 - \lambda_x)}, 0, \frac{1}{1 - k_{1G}(1 - \bar{\lambda})}\right). \end{aligned}$$

Plugging this solution in equation (24) allows to solve for k_{1V} and k_{1G} . Then we obtain the intercepts A_{0i} , $i \in \{V, G\}$, as

$$A_{0i} = \log \frac{k_{1i}}{1 - k_{1i}} - A_{1i}^\top \mu_X.$$

The risk premium of asset i is computed as

$$\text{RP}_i(t) = -\frac{1}{dt} \left\langle \frac{dM_t}{M_t}, \frac{dP_{i,t}}{P_{i,t}} \right\rangle.$$

□

B Summary Statistics

Table B.1 reports summary statistics on the growth forecast error and growth forecast of real GDP, real value portfolio earnings, and real growth portfolio earnings. The growth forecast error mean is close to zero for all three variables. This shows that analyst forecasts are accurate on average. However, earnings growth forecast errors feature substantial volatility. Indeed, the growth forecast error volatility of the value portfolio earnings is about four times larger than that of the growth portfolio earnings, and about 35 times larger than that of GDP. The mean growth forecast of GDP is 0.2 percentage points lower than that of the value portfolio earnings, and about two times lower than that of the growth portfolio earnings. For all three variables, the growth forecast volatility is about three times lower than the growth forecast error volatility.

Table B.1: GDP and Earnings Growth Summary Statistics.

This table reports summary statistics for GDP growth forecast error, GDP growth forecast, earnings growth forecast error and earnings growth forecast of the value portfolio, and earnings growth forecast error and earnings growth forecast of the growth portfolio. The data is in real terms and at the quarterly frequency from Q1:1984 to Q4:2022. All values are in raw quarterly units.

	GDP	Value Portfolio	Growth Portfolio
Growth forecast error mean	−0.0003	0.0004	0.0007
Growth forecast error volatility	0.0099	0.3445	0.0828
Growth forecast mean	0.0067	0.0087	0.0141
Growth forecast volatility	0.0030	0.1340	0.0318

C Additional Details on the Parameter Estimation

In the model, the forecast error (i.e., the difference between realized and forecasted cashflow growth rates) of the growth firm G and the forecast error of the value firm V have equal volatilities, whereas in the data the forecast error of the value portfolio is about four times more volatile than the forecast error of the growth portfolio. To make the model and the data

comparable, we scale both the realized and forecasted earnings growth rates of the value portfolio by the following multiplication factor: $\frac{vol(\text{growth portfolio forecast error})}{vol(\text{value portfolio forecast error})}$, where $vol(.)$ stands for volatility. This insures that the forecast error of the value portfolio and the forecast error of the growth portfolio have equal volatilities, as in the model. The demeaned GDP growth forecast represents the expected output growth component $\hat{x}$ in the model. The difference between the demeaned earnings growth forecast of the growth (resp., value) portfolio and the demeaned GDP growth forecast $\hat{x}$ represents the firm-specific expected growth component $\hat{x}_G$ (resp., $\hat{x}_V$) in the model. Observing forecast errors, the expected output growth component, the value-firm-specific expected growth component, and the growth-firm-specific expected growth component allows to estimate the parameters of the model by maximum likelihood. The maximum likelihood estimation is performed as follows. The parameters $(\sigma_C, \mu, \sigma_x, \lambda_x)$ are estimated using GDP forecast errors and GDP forecasts. The parameter σ_D is estimated using earnings forecast errors. The parameters $(\bar{\sigma}, \bar{\lambda}, p_V, p_G, \rho_V, \rho_G)$ are estimated using earnings forecasts of the growth portfolio, earnings forecasts of the value portfolio, and GDP forecasts. That is, the parameters $(\bar{\sigma}, \bar{\lambda}, p_V, p_G, \rho_V, \rho_G)$ are estimated using analyst forecasts only.

D Discussion on the Dynamics of the Cashflow-to-Output Ratios

This section discusses the dynamics of the cashflow-to-output ratios both in the model and in the data. The cashflow and output dynamics provided in Equations (1) and (4) imply that the cashflow-to-output ratios are not mean-reverting processes. Yet, these dynamics together with the parameter values provided in Section 4.1 imply that most of the simulated paths of the cashflow-to-output ratios feature a mean-reverting behavior.

We simulate 10,000 paths of the output, firm V cashflow, and firm G cashflow at the quarterly frequency over a 100-year horizon. The initial values are 13.7 for the output C , 0.5 for firm V cashflow D_V , and 0.5 for firm G cashflow D_G . That is, the initial total cashflow to output ratio is 7.3%, which corresponds to the average historical U.S. after-tax corporate profits to GDP ratio. The market cashflow is by definition: $D_m = D_V + D_G$.

To test whether the cashflow-to-output ratios are mean-reverting, we perform the following regression on each 100-year simulated path:

$$diff\left(\frac{D_{i,t+\Delta}}{C_{t+\Delta}}\right) = a + b \times \frac{D_{i,t}}{C_t} + \epsilon_{t+1}, \quad i \in \{V, G, m\}$$

where the differential operator satisfies $diff(x_{t+\Delta}) = x_{t+\Delta} - x_t$ and $\Delta = 1/4$. If the slope coefficient b is negative, then the cashflow-to-output ratio is mean-reverting.

The top panel of Table D.1 shows that the model-implied slope coefficient b is negative on average, over the 10,000 simulated paths. In addition, the slope coefficient b is negative on 84%, 87%, and 80% of the simulations for the value firm V cashflow-to-output ratio, growth firm G cashflow-to-output ratio, and market cashflow-to-output ratio, respectively. The bottom panel of Table D.1 shows that the slope coefficient b is negative and statistically significant at the 5% level for each cashflow-to-output ratio in the data. In summary, the mean-reverting properties of the cashflow-to-output ratios obtained in our calibrated model are in line with what is observed in the data.

Figure D.1: Distribution of Cashflow-to-Output Ratios at the 50-year Horizon.

This figure depicts the probability density functions (PDF) and cumulative distribution functions (CDF) of the model-implied cashflow-to-output ratios for the value firm V (top panels), the growth firm G (middle panels), and the market (bottom panels). The 10,000 simulated cashflow-to-output ratios are recorded at the 50-year horizon. The model parameters are provided in Section 4.1.

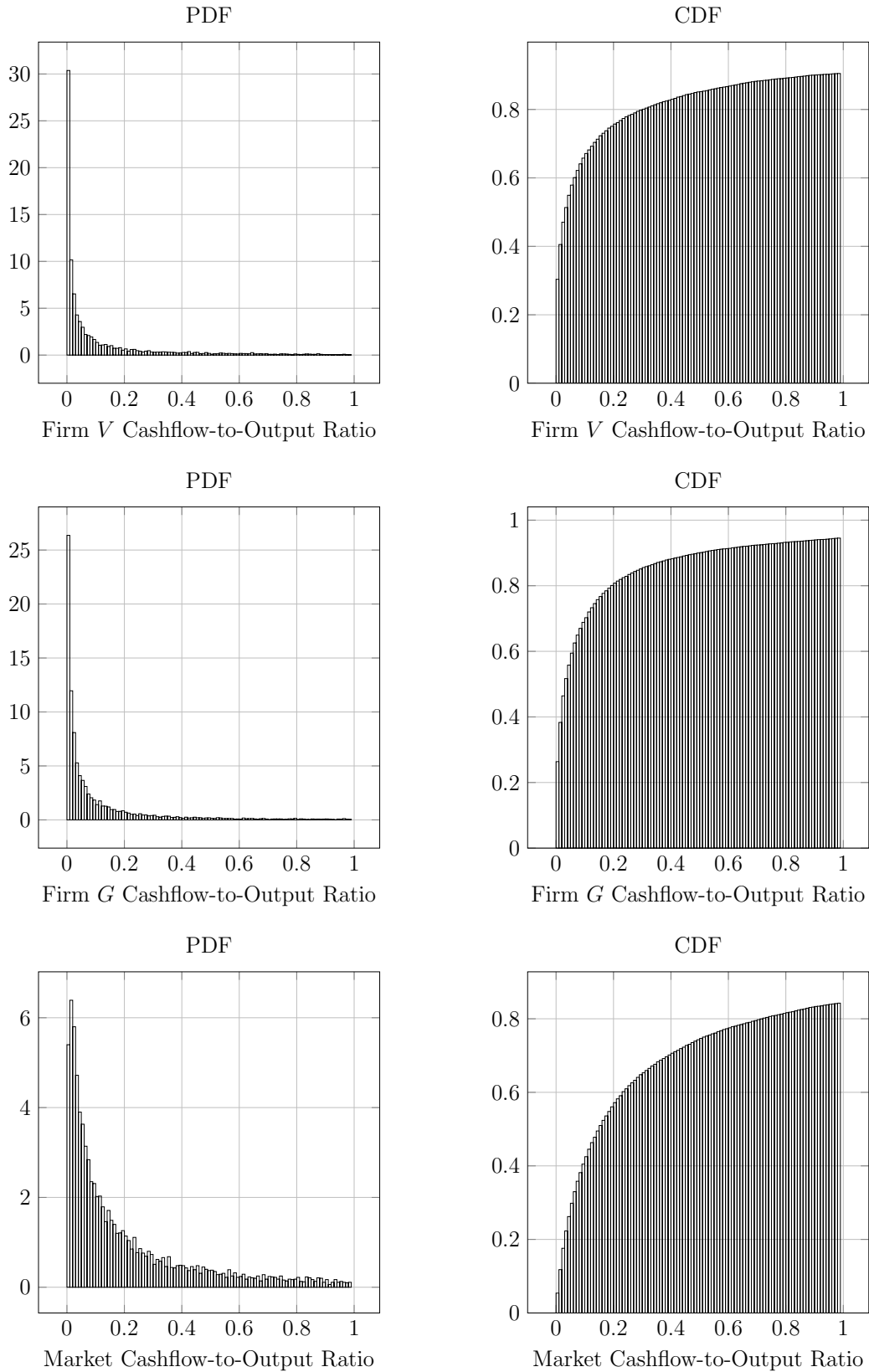


Figure D.2: Distribution of Cashflow-to-Output Ratios at the 100-year Horizon.

This figure depicts the probability density functions (PDF) and cumulative distribution functions (CDF) of the model-implied cashflow-to-output ratios for the value firm V (top panels), the growth firm G (middle panels), and the market (bottom panels). The 10,000 simulated cashflow-to-output ratios are recorded at the 100-year horizon. The model parameters are provided in Section 4.1.

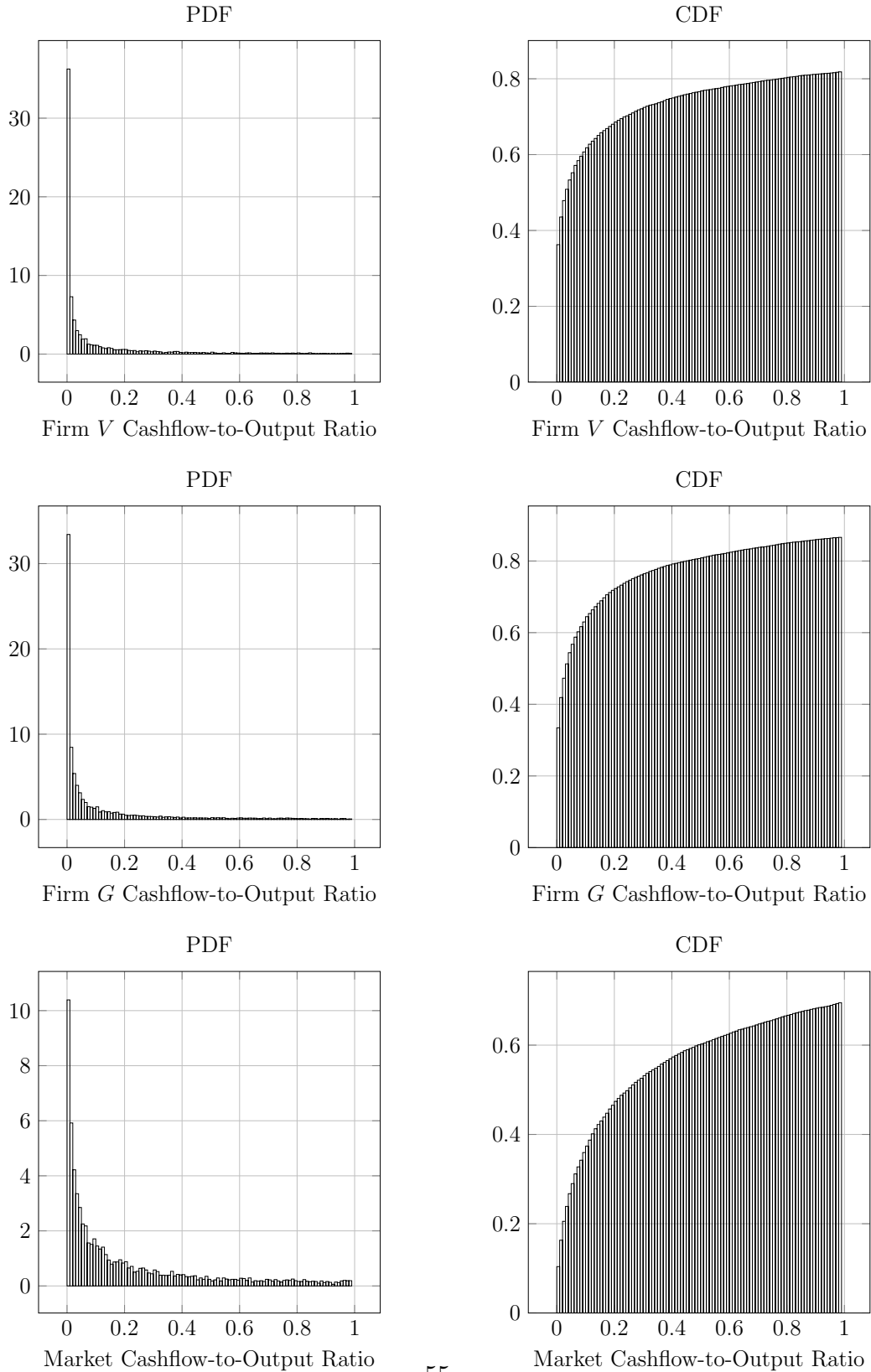


Table D.1: Mean-Reversion Tests on Cashflow-to-Output Ratios.

This table reports the output of the mean-reversion tests on cashflow-to-output ratios in the model (top panel) and in the data (bottom panel) for the value portfolio, growth portfolio, and market portfolio. In the model, 10,000 paths of the cashflow-to-output ratios are simulated at the quarterly frequency over a 100-year horizon. The model parameters are provided in Section 4.1. In the data, the output is the GDP, the cashflows of the value and growth portfolios are the total earnings of respectively value and growth firms in CRSP–IBES–Compustat, and the market cashflow is the total earnings of firms in CRSP–IBES–Compustat (see Section 3.2). As for the estimation, GDP and earnings data are in real terms and at the quarterly frequency from Q1:1984 to Q4:2022.

	<u>Model</u>		
	Value Firm V	Growth Firm G	Market m
$Mean(b)$	−0.0056	−0.0071	−0.0045
$Prob.(b < 0)$	0.8421	0.8704	0.8028

	<u>Data</u>		
	Value Portfolio	Growth Portfolio	Market Portfolio
b	−0.1352	−0.0358	−0.0506
$p\text{-value}(b)$	0.0136	0.0398	0.0230

Figures D.1 and D.2 depict the distributions of the cashflow-to-output ratios at the 50-year horizon and 100-year horizon, respectively. At both the 50-year and 100-year horizons, most of the probability mass of firms V and G cashflow-to-output ratios is concentrated between 0.02 and 0.15. As a result, most of the probability mass of the market cashflow-to-output ratio is concentrated between 0.04 and 0.30. These 50-year and 100-year values are reasonably close to the initial values. Indeed, the initial cashflow-to-output ratios are 0.0365 for firms V and G , and 0.073 for the market. At both the 50-year and 100-year horizons, the probabilities that the cashflow-to-output ratios become larger than one are reasonably low.